WELLINGTON MANAGEMENT®

2021 Climate Report

PREPARED IN ALIGNMENT WITH THE RECOMMENDATIONS OF THE TASK FORCE ON CLIMATE-RELATED FINANCIAL DISCLOSURES

Contents

Message from leadership

Climate change continues to be a key strategic focus for our firm. Apart from the COVID-19 pandemic, few issues so captured the world's attention in 2021 — a record year for many climate events. Our research into the market and economic effects of the physical and transition risks of climate change expanded significantly last year. This report shares updates on our approach to climate-related oversight, research, risk management, and engagement. We also provide an update on our Net Zero by 2050 commitment and how we are reducing our own carbon footprint.

Wellington endorsed the Task Force on Climate-related Financial Disclosures (TCFD) framework in 2017. The TCFD recommendations aligned with our view of climate change as a systemic risk to long-term financial performance. In 2018, we began a research collaboration with Woodwell Climate Research Center (Woodwell), one of the world's most respected independent climate research organizations, to bridge the gap between climate science and finance. The goal of this partnership is to deepen our appreciation of how physical climate variables, including heat, drought, wildfires, floods, hurricanes, water scarcity, and sea-level rise, will affect the securities we invest in on behalf of our clients.

In accordance with our participation in the Net Zero Asset Managers (NZAM) initiative, we are pleased to report US$436 billion client assets aligned with achieving net-zero emissions by 2050 or sooner. As of 31 March 2022, more than 32% of our total assets under management are now on track for net zero. Because we see this commitment as inextricably linked with our clients' long-term investment objectives, we evaluate client portfolios and strategies individually. Collaborating with clients, we expect to increase this commitment and will share our progress at significant milestones. We will report annually via our TCFD disclosures and submit a climate action plan to the Investor Agenda via its partner organizations.

Following due diligence throughout 2021, we announced in early 2022 a new climate-science research collaboration with the Joint Program on the Science and Policy of Global Change at the Massachusetts Institute of Technology (MIT Joint Program). We expect this alliance to strengthen our current research on the low-carbon transition, enhance our understanding of expected financial and economic impacts of transition pathways, and inform our engagements with companies and issuers. We believe an appreciation of climate science enhances our stewardship approach and the investment decisions we make on behalf of our clients.

We are expanding our climate research to explore adaptation and mitigation solutions across both public and private equity markets. We are broadening engagement efforts to encourage science-based target-setting and progress measurement at portfolio companies. And as a firm, we continue to search for new ways of reducing our corporate greenhouse gas (GHG) emissions.

We see all of these climate-related efforts as critical to our sustainability and core mission to drive excellence for our clients to positively impact millions of beneficiaries' lives.

Jean Hynes, CFA
Chief Executive Officer

Wendy Cromwell, CFA
Vice Chair and Head of Sustainable Investment

Chris Goolgasian, CFA, CPA, CAIA
Director of Climate Research

Introduction

Based on our research collaboration with Woodwell, we understand that the physical risks of climate change will become more severe and widespread over the next 10 to 30 years, with material implications for economies and capital markets. As society increasingly appreciates the severity of physical climate risk, we expect the transition to a low-carbon economy to accelerate, prompting changes in the regulatory environment and compelling companies to adjust their business models and become more energy efficient and resilient.

Climate dynamics may affect company fundamentals in a variety of ways. Higher costs in the form of capital expenditures, insurance, and carbon taxes are all possible, as are business disruptions from supply-chain interruptions, regulatory scrutiny, litigation, and shifting consumer preferences. For some industries, climate change also increases the potential for impaired and/or stranded assets. In our view, markets are not yet fully pricing climate risks — nor the potential opportunities for companies that can successfully adapt to or mitigate these risks. Historical bias (failing to appreciate that climate events will occur at a faster pace than in the past) and behavioral bias (failing to take action amid the depressing climate outlook) are two key reasons for persistent mispricing.

As fiduciaries of our clients' assets, we take a pragmatic approach, viewing climate change through a financial-market lens. We strive to better understand the economic risks, advocate for standard disclosures, and enable capital to find its way to climate leaders. This work supports our objective of generating competitive, nonconcessionary investment returns and positive client outcomes.

One of Wellington's key strategic initiatives is to integrate the assessment of sustainability-related risks, including climate considerations, into our investment decision making, as relevant to each investment approach. The firm has mobilized around this opportunity and taken steps to ensure we exemplify the behaviors we seek from the portfolio companies we own on behalf of our clients. To that end, we increasingly focus on corporate sustainability. WellSustain, our firmwide sustainability platform, formalizes the incorporation of sustainability into our operations and infrastructure. WellSustain centers on environment, community, and culture. Reducing our firm's carbon footprint is our main environmental objective.

We are pleased to share our third annual climate report. Aligned with the TCFD's recommendations, this report describes our data-driven processes for monitoring, evaluating, and managing material climate-related risks and opportunities. We also outline the processes, metrics, and output of our climate disclosures, which will evolve over time as more and better data becomes available. To learn more about our overall approach to sustainable investing, please see our Sustainability Report, available on our website.

ABOUT WELLINGTON'S CLIMATE CAPABILITIES

Wellington Management is one of the world's largest independent investment management firms, serving as a trusted adviser to over 2,400 institutional and private clients located in more than 60 countries and actively managing over US$1.4 trillion of assets, as of 31 December 2021. With capabilities covering most segments of the global capital markets, our focus is on investment management on behalf of our clients. As of year-end 2021, our climate platform represented US$28 billion in AUM across climate-solution strategies, low-carbon/Paris-aligned approaches, and climate-aware strategies.

Investment teams draw from extensive proprietary research and benefit from a collaborative culture encouraging independent thought and rigorous debate. An ongoing, science-led climate-research agenda informs how we evaluate physical and transition risks and consider opportunities for growth. The Climate Research Team helps translate scientific data and projections into investment data, tools, and insights, which they make available to investors across the firm. We aim to ensure that climate data helps inform investment decisions across asset classes and geographies. We believe this work allows our investment teams to better assess climate-risk exposures and climate-related return opportunities by improving their ability to anticipate climate impacts in the context of their investment objective and time horizon, and as relevant to each investment team's investment philosophy. Investors at the firm often share these differentiated insights in engagements with companies and issuers, raising awareness of climate risks across market sectors and geographic regions.

EVOLVING CLIMATE LEADERSHIP

We launched our first climate-focused equity approach in 2007 and have been researching the effects of climate change on investments for decades via our Future Themes initiative. Future Themes is a firmwide special-research project conducted every five to seven years, focused on secular themes that may impact markets long term. This research has included climate and environmental topics since 1984. Our collaboration with Woodwell in 2018 crystallized our commitment to becoming a leader on climate change in our industry.

2017

Launched systematic strategy with low-carbon focus

Signed statement of support for the TCFD and published first TCFD report

2018

Began Woodwell Climate Research Center collaboration

Supported Transition Pathway Initiative (TPI)

2019

Formalized Climate Research Team

Director of Climate Research named to CFTC[1] Climate-Related Market Risk Subcommittee and contributed to Managing Climate Risk in the U.S. Financial System report

Published Physical Risks of Climate Change (P-ROCC) disclosure framework

Joined CDP (formerly Carbon Disclosure Project)

Joined Ceres Investor Network on Climate Risk and Sustainability

2020

Launched climate-resilience strategy

Became a founding member of NZAM

Joined Climate Action 100+

Joined the Institutional Investors Group on Climate Change (IIGCC)

2021

Published P-ROCC 2.0: Call for location data

Head of Sustainable Investment joined NZAM Advisory Group

Launched climate-focused private equity and alternatives strategies

Multiple colleagues joined working groups within Glasgow Financial Alliance for Net Zero (GFANZ) and Paris Aligned Investment Initiative (PAII)

Climate transition-risk analyst joined the PAII's Top-down Target Setting subgroup as co-leader

Conducted 540 company physical climate-risk assessments

Developed a climate-adjusted strategic asset allocation framework

Launched proprietary net-zero dashboard, enabling investment teams to analyze climate transition risks

Liaised with the US Securities and Exchange Commission (SEC) to advocate for climate disclosure standards

Commented on the US Environmental Protection Agency's (EPA's) proposed Oil and Natural Gas Emissions Guidelines

Entered into a power purchase agreement (PPA) with Enel Green Power to match 100% of US office and employees' home electricity consumption

Announced a long-term 105,000 sq. ft. lease, beginning in 2023, to a LEED Zero Carbon Certification building in a Boston suburb

Installed 1,100 LED lights in our London offices to help reduce our electricity usage

2022

Entered research collaboration with MIT Joint Program studying climate transition risks

Joined Global Real Estate Engagement Network (GREEN)

Committed to become carbon neutral in our operations by year end

1 US Commodity Futures Trading Commission.

Section 1: Oversight and governance

During 2021, we strengthened the governance for our sustainability practice to ensure we can best support our objectives across our investment, client service, and infrastructure platforms. The following is a detailed look at our structure.

SUSTAINABILITY GOVERNANCE AT A GLANCE

	Oversight	Management & accountability	Research & assessment
Firmwide governance			
Sustainable Investment (SI) Governance Committee	✗		
SI Management Team		✗	
Firmwide stewardship			
Investment Stewardship Committee	✗	✗	
SI Product Integrity Working Group		✗	✗
Global Exclusions Working Group		✗	✗
Portfolio integration			
Climate Research Team			✗
ESG Philosophy & Process panels	✗		
Portfolio oversight			
Line management	✗		
Investment Product and Fund Strategies	✗	✗	
Investment Review Groups	✗		

FIRMWIDE GOVERNANCE

These are still relatively early days for sustainability in our industry and, like our peers, we continue to learn as we go. Regulation is changing rapidly. Our employees are seeking to understand sustainability issues — including climate change — more extensively. To continue delivering excellence for our clients, we have developed several forums designed to optimize cross-functional decision making. With the creation of our Sustainable Investment (SI) Governance Committee and SI Management Team, we have evolved our governance model to deepen firmwide sustainability knowledge, formalize responsibilities, improve communication, ensure ongoing collaboration, and facilitate efficient decision making.

SI GOVERNANCE COMMITTEE (SIGC)

Purpose: Oversee the firm's sustainability efforts; champion the sustainable investment vision and support sustainability teams firmwide to help ensure their success

Membership: Wellington's CEO and senior leaders from across the firm, including from our Executive, Operating, and Compensation Committees

Meeting frequency: Monthly

Sample activities: Establish/renew key external partnerships; approve major industry initiatives; endorse resource recommendations across platforms; approve policy mandates with broad, firmwide consequences

SI MANAGEMENT TEAM (SIMT)

Purpose: Determine and execute the firm's overall SI vision and strategy to meet evolving investment, client, and regulatory imperatives

Membership: SI leaders from our investment, client, and infrastructure platforms

Meeting frequency: Twice weekly

Sample activities: Recommend and prioritize resource needs; decide industry-wide working group participation; identify resources for SI initiatives; identify and solve for gaps, redundancies, or inconsistencies arising from our decentralized SI management model

FIRMWIDE STEWARDSHIP

Our goal is to familiarize ourselves with our clients' stewardship and investment policies and discuss how we can adhere to their requirements. For any policies we cannot adhere to, we proactively communicate our position. Once we begin managing assets on a client's behalf, we want to be sure our management aligns with a mutually agreed-upon approach.

INVESTMENT STEWARDSHIP COMMITTEE (ISC)

Purpose: Set the strategic direction for firmwide stewardship, focusing on proxy voting, engagement, and portfolio exclusions

Membership: Senior-level and experienced professionals from portfolio management, investment research, sustainable investment, corporate sustainability, relationship management, and legal and enterprise risk

Meeting frequency: Every other month

Sample activities: ISC members engage in dialogue and debate on stewardship policy. They also approve guidelines and practices. The committee is empowered to:

- Set and approve our proxy voting policies and procedures, conflicts-of-interest policy, and annual voting guidelines
- Oversee proxy votes, with a focus on key stewardship issues and evolving best practice
- Set and approve our engagement policy
- Monitor engagement practices and steer engagement priorities
- Advise firmwide on engagement and stewardship matters, including escalation and conflicts
- Confirm that we satisfy our stewardship code responsibilities
- Ensure we are accountable and authentic in our stewardship commitments
- Identify tools and information to support investors in their stewardship decisions
- Oversee the Global Exclusions Working Group

SI PRODUCT INTEGRITY WORKING GROUP

Purpose: Ensure that Wellington's sustainable investment products, funds, and client solutions have high integrity and are operationally scalable, in alignment with our overall SI strategy and global regulatory and commercial standards

Membership: Experienced professionals from our investment, client, and infrastructure platforms, including colleagues focused on legal, compliance, and investment-guideline monitoring

Meeting frequency: Weekly

Sample activities: Evaluate new SI products and recommend whether they should be offered and marketed to clients, considered for incubation, or not pursued in their current form; evaluate custom SI client solutions and evolution of existing products and funds; consider investment integrity, commercial potential, legal implications, and operational complexity in the context of the overall suite of sustainable investment product offerings and capabilities

GLOBAL EXCLUSIONS WORKING GROUP

Purpose: Develop, monitor, and evolve frameworks for exclusions and enhanced engagement protocols related to economic activities commonly incorporated into portfolios. These protocols are informed by Wellington research and are used by select Wellington sponsored funds and Wellington clients who elect to adopt this policy upon request. This group reports to the ISC.

Membership: Experienced professionals from our investment, client, and infrastructure platforms

Meeting frequency: Twice monthly

Sample activities: Assess client and market expectations related to business activities our clients seek to avoid (e.g., antipersonnel mines) or promote (e.g., low-carbon transition, responsible business practices); articulate internal and external communication plans regarding enhanced engagement protocols, escalation processes, or exclusions; evaluate new and existing categories for exclusions or enhanced engagement as needed

PORTFOLIO INTEGRATION

Wellington fosters an open, collaborative culture in which our investors share research and challenge one another to sharpen decisions, uncover blind spots, and manage risk. Our assessment of sustainability-related risks is integral to this culture. We combine multidisciplinary resources — including equity and credit analysts, ESG research analysts, climate researchers, data scientists, and macro analysts — with portfolio teams. We refer to this structure as a community of investment boutiques, each with its own philosophy and process (P&P) to inform investment decisions. We believe this model helps us attract talented, independently minded investors and align their efforts with client outcomes. We also find diversity of thought strengthens internal debate and, as a result, decision making on behalf of our clients.

CLIMATE RESEARCH TEAM

Purpose: Research climate physical and transition risks in collaboration with our external climate-science partners; apply and communicate climate-research insights to help investment teams understand the potential impacts of climate change on markets and economies and make more informed decisions; help identify and assess investment risks and opportunities across sectors, geographies, and asset classes

Membership: Director of Climate Research, two climate transition-risk analysts, one climate physical-risk analyst, one geospatial analyst, four rotational early-career investors

Meeting frequency: Twice monthly broad climate meetings with investors to present research from Woodwell and the MIT Joint Program; quarterly meetings with the broader Wellington investment community summarizing research and investment insights; informal meetings as needed with investors around the firm

Sample activities: Host meetings to facilitate dialogue among investors and climate-science partners; publish notes on our firmwide collaborative research-technology platform; present quarterly findings to the firm; build investor tools for portfolio analysis and risk management; participate in Wellington's early-career investor development program; partner with ESG research analysts to integrate climate insights into company-specific research and engagements; source scientific and climate-related company data

ESG PHILOSOPHY AND PROCESS PANELS

Purpose: Review the philosophy and process of all investment teams, with a specific focus on authentic ESG integration. Discuss how each investment team considers material ESG factors in investment approaches, including climate risks, and how teams incorporate those factors into their overall investment process.

Membership: Senior investment professionals, sustainability experts, line management

Meeting frequency: Regular and ongoing (with a goal of reviewing each strategy once per year)

Sample activities: Review topics such as ESG integration into the investment process; provide written feedback to investors highlighting both strengths and areas for further work

PORTFOLIO OVERSIGHT

LINE MANAGEMENT

Purpose: Oversee portfolio managers and investment groups

Membership: Senior line managers across the investment platform

Meeting frequency: Regular and ongoing

Sample activities: Assist with talent resourcing and development needs, opine on advancement and compensation decisions, and meet regularly with portfolio managers to provide feedback and support. Line managers also collect and provide feedback to portfolio managers from the firm's independent oversight groups, including Investment Product and Fund Strategies (IPFS), ESG philosophy and process panels, and investment review groups.

INVESTMENT PRODUCT AND FUND STRATEGIES (IPFS)

Purpose: Provide independent oversight of portfolios with a focus on investment integrity, specifically to assess whether each portfolio's holdings are consistent with clients' investment objectives and risk tolerance

Membership: Investment directors, specialists, and analysts, dedicated to specific strategies. IPFS is independent from each portfolio management team, with different reporting lines and the ability to escalate concerns to line management.

Meeting frequency: Regular and ongoing

Sample activities: Meet with portfolio managers who have designated climate risks and opportunities as part of their decision-making process; review portfolio-level climate data, climate-related voting, engagement tracking, and other relevant ESG and stewardship factors; work with investment teams and clients to identify and disclose sustainability risks; join portfolio managers' investment discussions and firmwide peer review groups

INVESTMENT REVIEW GROUPS

Purpose: Provide advice and guidance to foster portfolio manager development; offer insights into line management's fiduciary oversight; conduct formal and informal reviews of investment teams' P&P, investment performance, and other pertinent topics

Membership: Senior investment professionals with relevant market experience

Meeting frequency: Regular and ongoing (with a goal of reviewing each strategy once per year)

Sample activities: Ask portfolio managers how they integrate climate risks and opportunities into their P&P; provide detailed feedback with line management and other review groups

INCENTIVE ALIGNMENT

A key structural support of our collaborative environment is our performance review and compensation process. We structure our compensation program to incentivize longer-term performance measurement, which helps align climate time horizons and investor time horizons. Our clients often have decades-long obligations to their beneficiaries. Because climate-related risks and opportunities are present in the market today and will be relevant for decades to come, our fiduciary duty to clients obligates us to incorporate near- and longer-term perspectives.

As part of the investment mosaic, and as relevant to each investment team's investment philosophy, climate insights help us pursue our goal of delivering competitive risk-adjusted returns in the portfolios we manage on behalf of our clients. Performance evaluations of our research analysts include an assessment of whether analysts demonstrate depth of knowledge of the key ESG issues in their sector and company views. In addition to compensation based on the quality and depth of their research, analysts are eligible to receive discretionary bonuses according to how successfully investors implement their recommendations in client portfolios across the firm.

Section 2: Climate research, tools, and risk management

We believe climate change poses material risks for companies, economies, and society — and therefore, for our clients' investment portfolios. Understanding climate factors, particularly those material to specific industries or companies, informs long-term investment decisions, promotes leadership on best practices, and enables us to engage more effectively on our clients' behalf. Extensive bottom-up research and close collaboration among our investment teams, ESG Research, and Climate Research teams informs our approach. Each investment team is responsible for climate-risk management within their portfolios. Each investment team has its own investment philosophy and process ('P&P'); the extent to which climate research impacts investment decision making will vary depending on the P&P. Investors have proprietary tools at their disposal to help them integrate climate insights into their decision-making process.

These resources bolster our risk management and underpin a common research philosophy centered on:

- **Rigor**: All research used in investment decision making should be precise and held to scientific peer-reviewable standards
- **Application**: Capital market insights from climate research can help inform active investment decisions
- **Evolution**: As climate and ESG data improves and as our research progresses, so too will our insights and conclusions

While the scope of our climate research is broad, our ability to quantitatively measure climate risks depends on data availability. In 2021, we committed significant resources to enhance data availability and develop investor tools to facilitate better measurement of potential climate risks. Our physical climate research can help investors assess the risk exposure of securities across most global regions and asset classes, so long as physical locations related to an investment are known. The scope of our transition-risk research and issuer evaluations have focused thus far on corporate instruments, as emissions data and standards are most developed in this asset class. We expect to expand our transition-risk research and security-level evaluations to other asset classes as emissions disclosure improves.

COLLABORATIVE CLIMATE INTEGRATION

Wellington has long believed that an open, collaborative culture benefits our clients and strengthens our firm. By sharing research and challenging one another's views, our investors and analysts create a diverse marketplace of ideas. The ESG Research and Climate Research teams are integral to this ecosystem.

The structure of Wellington's integrated research functions lends itself well to analyzing the impact of climate change on economies, industries, and companies. Our ESG, equity, and credit analysts coordinate their views on the value of securities in a given industry, while our climate and macro analysts contribute industry insights via thematic lenses. The result is a holistic research process that helps our investors appreciate the material and interconnected nature of climate change. We expect to expand our team of climate-dedicated investors to identify opportunities in private equity markets.

FOUNDATIONAL GOAL: UNDERSTAND THE ECONOMIC IMPACTS OF CLIMATE CHANGE

Our ultimate goal is to recognize and address how and where climate change may affect capital markets and economies over time. Our Climate Research Team drives these efforts and serves as a key research resource for investors across the firm, providing insights, data, and tools for use as inputs into each investor's research mosaic. Our research and tools cover both physical climate change and the transition to a low-carbon economy. This work facilitates investment research, engagement meetings, and investment-opportunity discovery.

PHYSICAL CLIMATE-RISK RESEARCH AND INVESTOR TOOLS

During our four-year collaboration with Woodwell, we have been researching heat, drought, wildfires, floods, hurricanes, water scarcity, and sea-level rise. Our objective is to bridge the gap between climate science and finance to better understand the potential impact of these variables on capital markets, identify potential asset mispricings, and produce financially relevant insights.

Effective climate-aware investment decision making depends on access to reliable data and resources to integrate that data into risk analysis. Climate science is complex, however, and climate models do not easily translate into financial models and projections. We believe this mismatch results in asset-pricing dislocations.

Our Woodwell partnership has provided the data and resources necessary to enhance our physical-climate-risk analysis and integrate this research into our investment process. The Climate Research Team regularly engages with investment teams across the firm to answer questions and solicit feedback. Based on this collective foundation of climate science and investment knowledge, we have also developed several proprietary tools available to all Wellington investment teams. Details about these processes and tools follow.

REGULAR RESEARCH AND INVESTMENT MEETINGS

Members of our ESG Research Team, Climate Research Team, and climate-dedicated investment teams join sector-focused analysts and diversified portfolio managers to debate the impacts of physical and transition risks on the securities in their opportunity set. These teams participate in Wellington's twice-daily Morning Meeting (to accommodate multiple global time zones), in which multidisciplinary investment teams share perspectives.

In addition, the Climate Research Team and climate scientists from Woodwell regularly host informal meetings to review Woodwell's climate-research findings. The Climate Research Team steers research topics along market-relevant lines. On a quarterly basis, the team summarizes its research and investment insights and presents these to the broader Wellington investment community.

CLIMATE EXPOSURE RISK APPLICATION (CERA)

Working with Woodwell, we have developed an innovative physical-climate-risk assessment tool. Using integrated spatial finance, CERA displays geospatial maps, which, when overlaid with climate data, enable us to visualize and quantify physical climate risks for a wide variety of securities and real assets (**Figure 1**). CERA allows investors to isolate or combine views of seven key climate factors: heat, drought, floods, hurricanes, wildfires, water scarcity, and sea-level rise. The tool overlays capital market insights onto regional maps, enabling portfolio managers and investment teams to identify "hot spots" warranting further fundamental research for each asset class and region of interest. The Climate Research Team applies the information and insights from CERA, along with company disclosures, to issuer analysis. To date, the team has evaluated 1,600 companies for their physical risks, based on materiality of exposure and strength of risk-management practices.

Figure 1

Screenshot from CERA

Source: Wellington Management

PHYSICAL CLIMATE-RISK RATINGS AND CLIMATE PORTFOLIO REVIEWS (CPRS)

The Climate Research Team conducts comprehensive physical-risk assessments of individual issuers, prioritized according to the needs of investment teams. The team can aggregate these assessments in a CPR to highlight portfolio exposures with material physical climate-risk exposure. The Climate Research Team uses natural language processing (NLP) and manual reviews to analyze annual reports, disclosures, CDP disclosures, earnings transcripts, and investor presentations. This information helps our investors assess the materiality of a company's climate risk and current management of climate exposure, which we pair with our geospatial analysis, using CERA. The result is a climate-risk materiality assessment, captured via a physical-risk rating, and specific ideas for engagement topics. Investment teams across Wellington can then use this research to engage with company management teams directly on the topic of physical-risk management.

In recent years, the Climate Research Team has applied this process to evaluate physical climate risks for the holdings within 38 portfolios across equity and fixed income securities. In all, we have reviewed 1,600 companies globally, covering 43% of the names and 72% of the market value of the S&P 500 Index, and 20% of the names and 54% of the market value of the MSCI ACWI (**Figure 2**). The team aims to expand its research coverage further.

SUSTAINABILITY VIEW

This company-specific dashboard supports the investment teams' ESG and climate integration efforts, stimulates dialogue, and informs engagement prioritization by highlighting material topics for discussion. Sustainability View is part of Mosaic, our broader collaborative research-technology platform. This dashboard includes a variety of sustainable investment data and research including climate-specific information:

- Climate-informed ESG Fundamental (ESG-F) ratings, with rationale and access to underlying quantitative data inputs
- Physical climate-risk materiality assessments and engagement suggestions, drawn from CPRs, with supporting commentary
- GHG emissions-intensity data (for Scopes 1, 2, and 3) relative to industry[2]

2 Scope 1 emissions are released during regular operations at sites directly controlled by the company, including the use of natural gas for heating, generators, and company vehicles. Scope 2 emissions are indirectly generated from electricity and heat used to power the company's sites, so Scope 2 emissions depend upon the generation mix of the electricity grid at different sites. Scope 3 emissions are indirectly generated throughout the value chain, from upstream activities such as purchased goods and business travel and downstream activities such as the use of sold products.

Figure 2

Accounting of internal climate-risk ratings

Reviewed names by climate-risk rating	S&P 500	ACWI
Total % of names in index with CPR	42.60%	20.01%
Low/No risk identified	25.00%	21.59%
Potential risk identified	37.74%	36.93%
Material risk identified	37.26%	41.48%

Reviewed market value by climate-risk rating	S&P 500	ACWI
Total % of Market Value in index with CPR	72.08%	54.45%
Low/No risk identified	24.85%	16.00%
Potential risk identified	29.47%	23.02%
Material risk identified	17.76%	15.43%

Source: Wellington Management. Risk assessments as of 31 December 2021. Analysis is based on company disclosures as well as proprietary research done by Wellington Management's Climate Research Team. Coverage to date has been focused on corporate entities only and research on the broad global equity and corporate investment universe is incomplete. Ratings reflect a fundamental assessment of potential physical climate risk, based on our understanding of climate change projections (as supplied to Wellington by Woodwell Climate Research Center) and our research to identify key addresses associated with the company's operations. For companies researched thus far, there can be no assurance all addresses currently associated with company operations were captured, as most addresses are not publicly disclosed by companies today.

Drawing on Woodwell's climate projections and our proprietary geospatial mapping tool, we evaluate the potential impact of key climate variables including heat, drought, water scarcity, wildfire, floods, hurricanes, and sea-level rise for each known company address. These climate projections are subject to many assumptions as specified by the climate scientists at Woodwell. Projections vary over different time periods and are subject to change.

Materiality assessments are communicated via Low/No risk, Potential risk, and Material risk (defined below) and reflect potential physical risk identified by Wellington's Climate Research Team as of a specific time. If the company's disclosure of location data is more general than provincial or state level and we are unable to find additional information from supplementary sources, it would be insufficient to produce a rating based on our criteria. In this case, we would look at qualitative factors to assign a rating and flag the rationale with the statement (Qualitative assessment applied due to insufficient location data). Importantly, these ratings do not reflect our assessment of how company management teams plan to manage these risks over time. Ratings are not intended to signal buy or sell investment decisions. Ratings should only be interpreted as a potential input into an investment team's research mosaic. Other investment teams at Wellington Management may disagree with the Climate Research Team's materiality assessment.

Material risk identified: Typically defined as a company with greater than 10% of its key locations at risk, or greater than 5% of revenue at risk from climate-related events.

Potential risk identified: Typically defined as a company with some level of climate risk to its key inputs or critical facilities (1% -10%, or below 5% of revenue at risk), or a company with more than a 30% net property, plants, and equipment (PP&E) to total assets ratio.

Low/No risk identified: Typically defined as a company with little to no level of climate risk due to either its locations, diversification, the nature of its business, or the industry/sector in which it operates.

CLIMATE-ADJUSTED YIELD CALCULATOR

The Climate-adjusted Yield Calculator, designed for our fixed income teams, helps investors evaluate bonds with exposure to fixed locations. This tool integrates the probability of physical climate events (acute or chronic) onto a security's spread in stochastic fashion. The result is an adjusted yield that integrates the probability of a climate event, allowing investors to appreciate if the price of a bond accurately reflects climate risks.

CLIMATE TRANSITION-RISK RESEARCH AND INVESTOR TOOLS

Climate transition risks affect the ecosystem in which companies operate. Businesses effectively mitigating transition risks may accrue competitive advantages and see their cost of capital decline. Because transition factors vary by region and sector, we analyze them through various lenses, aiming to identify companies bolstering their readiness and engaging with those failing to do so. Our risk analysis entails disclosure reviews of a company's transition-related initiatives, which we post on our internal CDP dashboard. These assessments inform our ESG Research Team's ratings, also available to investment teams. Climate considerations — a key input to our ESG ratings — are informed by third-party data and through engagement. Our ESG ratings also include a proprietary materiality assessment.

We are in the process of developing a quantitatively derived transition alignment rating for corporate securities, consistent with the parameters of the Institutional Investors Group on Climate Change's (IIGCC's) Net Zero Investment Framework. We will continue to complement our quantitative assessments with issuer engagement and qualitative assessments of the credibility and feasibility of each company's transition-risk strategy.

COMPANY CARBON DASHBOARD

In 2021, we developed a new tool to compare companies more easily with peers and industry intensity averages. This dashboard includes the historical trend of intensity figures for Scopes 1 and 2 and Scopes 1, 2, and 3, leveraging estimated Scope 3 datasets (**Figure 3**). We display multiple data sources to demonstrate the difference in estimation methodologies and underscore the importance of encouraging company-specific disclosures. Analysts can look across their coverage universe to identify how business-strategy differences relate to carbon output. They can also identify lagging companies that may require more robust transition strategies and related capital expenditures to catch up to peers.

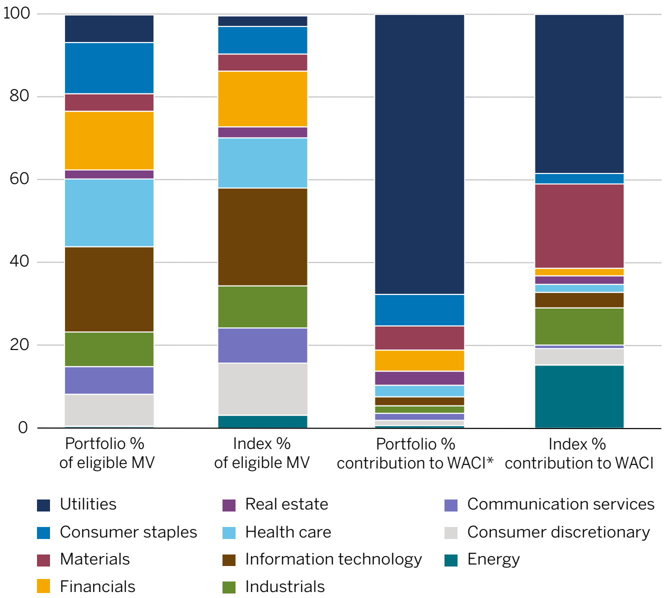
Figure 3

Carbon emissions data

SUMMARY

	Carbon emissions TCO2e	Carbon emissions/ US$M TCO2e/ US$M invested	WACI TCO2e/ US$M sales
Portfolio	46,155.6	20.2	96.4
Index	97,338.2	42.5	129.8

SECTOR CARBON INTENSITY



* Weighted average carbon intensity. For illustrative purposes only.
Sources: MSCI, Wellington Management

CARBON EMISSIONS DATA COVERAGE

% market value (MV) eligible for carbon data	% of eligible MV with carbon data				Without carbon data
	Adjusted	Estimated	Reported	Total	Total
99.5	6.2	15.3	78.5	100.0	0.0
100.0	3.8	12.9	83.1	99.8	0.2

	% of eligible MV		Contribution to WACI	
	Portfolio	Index	Portfolio	Index
Utilities	6.7	2.6	67.6	38.4
Consumer staples	12.3	6.7	7.7	2.5
Materials	4.3	4.1	5.8	20.4
Financials	14.2	13.4	5.0	1.8
Real estate	2.1	2.7	3.5	2.1
Health care	16.4	12.1	2.7	1.9
Information technology	20.6	23.6	2.2	3.8
Industrials	8.4	10.2	1.8	8.9
Communication services	6.7	8.5	1.7	0.9
Consumer discretionary	7.7	12.6	1.3	4.0
Energy	0.4	3.2	0.7	15.3
Not covered	0.0	0.2	0.0	0.0
Total	**100.0**	**100.0**	**100.0**	**100.0**
Not eligible	0.5	0.0		

For illustrative purposes only. Sources: MSCI, Wellington Management

NET-ZERO DASHBOARD

In 2021, our Climate Research Team developed a proprietary net-zero dashboard that allows us to monitor progress toward our net-zero milestones. The dashboard shows top-down progress, based on historical and projected portfolio-level weighted average carbon intensity (WACI). (See Section 4 for more information on WACI.) The dashboard also highlights bottom-up progress, measuring portfolio exposure to companies that have committed to or have set science-based targets (SBTs). Finally, this tool illustrates security-level exposures, including top contributors to each portfolio's carbon footprint. It suggests priority candidates for engagement on transition risk, based on their contributions to portfolio WACI and lack of alignment with the Science Based Targets initiative (SBTi) (**Figure 4**).[3]

Figure 4

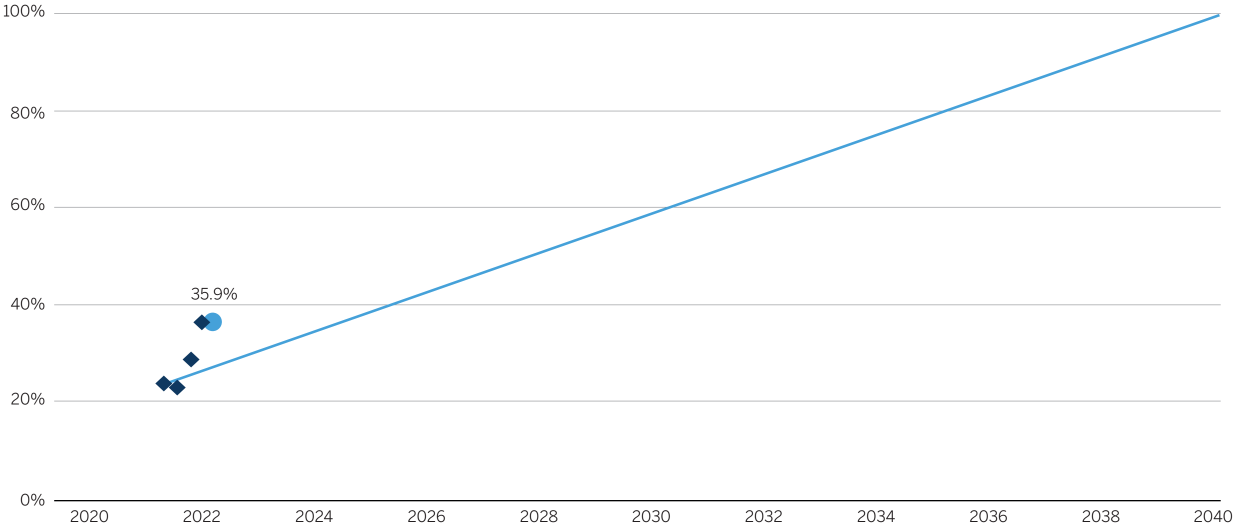
Exposure to companies with SBTs

SUMMARY

	% of eligible MV with SBTs		# of issuers with SBTs		Contribution to WACI %	
	Portfolio %	**Index %**	**Portfolio**	**Index**	**Portfolio %**	**Index %**
Total (Target set or Committed)	43.7	41.1	83	499	27.6	28.1
Targets set	35.9	28.8	67	348	22.4	18.6
Committed	7.8	12.4	16	151	5.2	9.4

GLIDEPATH

Required exposure to meet net-zero commitment using SBTs



For illustrative purposes only. Sources: MSCI, Wellington Management

NEW PARTNERSHIP WITH MIT JOINT PROGRAM

To bolster our transition-risk research, in January 2022, Wellington and the Joint Program on the Science and Policy of Global Change at the Massachusetts Institute of Technology announced the formation of a climate change research collaboration. The alliance allows us to enhance our research on the transition to a low-carbon economy, particularly with regards to our comprehension of the expected financial impacts of various transition pathways on industries and economies. We also expect this research will allow us to deepen our decarbonization engagement practices.

The MIT Joint Program's integrated team of natural and social scientists aims to provide Wellington's investment teams with comprehensive climate change projections under various environmental, economic, and policy scenarios. The objective of this research is to outline decarbonization pathways for corporate operations, supply chains, and products, while also assessing their potential economic impacts. Wellington's investment teams plan to integrate these transition-risk findings into their ongoing fundamental research, in conjunction with physical-risk findings from Woodwell.

3 SBTi is a partnership between the CDP, the United Nations Global Compact, World Resources Institute (WRI), and the World Wide Fund for Nature (WWF). More information on the initiative can be found at sciencebasedtargets.org.

OUR NET-ZERO COMMITMENT

In December 2020, we became a founding signatory of the Net Zero Asset Managers (NZAM) initiative. Our head of Sustainable Investment now sits on the NZAM Advisory Group, helping to shape the initiative and working with other asset managers to broaden the signatory group and encourage wider adoption. We have met with more than 20 potential and recent NZAM asset-manager signatories to share our journey, offer advice, and answer questions.

We ground our leadership in NZAM on our primary objective to achieve clients' investment goals and our secondary objective to demonstrate leadership on climate integration. We have also been able to enhance our engagements and consultative partnerships with clients and industry leaders. Through NZAM, we commit to the following:

- Partner with asset-owner clients on their decarbonization goals, consistent with achieving net-zero emissions by 2050 or sooner
- Set an interim target for the proportion of assets to be managed in line with this goal
- Review the interim assets-under-management (AUM) target at least every five years with the aim of increasing the proportion of assets covered, until 100% are net zero
- For assets committed to net zero, set an interim 50% emissions reduction target by 2030

As demand increases and more of our clients make net-zero commitments, and as better data and methodologies become available, we expect committed assets under management to increase over time.

NZAM PROGRESS UPDATE

During 2021, we built a foundation for our investment teams and clients to establish net-zero objectives and measure progress with an emphasis on:

- Discussing net-zero implementation options and measurement approaches with clients
- Engaging with companies to promote science-based targets as part of credible climate transition plans
- Incorporating additional carbon-related data into proprietary investment research tools to foster increased transition-risk awareness, measurement, and accountability
- Defining measurable net-zero implementation plans for 59 investment strategies across equities and corporate fixed income
- Actively serving on the net-zero advisory board and participating in multiple implementation-oriented working groups

As a result of these and other efforts, we have a commitment of US$436 billion in client assets aligned with achieving net-zero emissions by 2050 or sooner, as of 31 March 2022. Because we see our net-zero commitment and clients' investment objectives as inextricably linked, we assess clients' investment portfolios and investment strategies one by one. Our initial commitment represents assets held in accounts in which:

- A client-approved decarbonization objective consistent with net zero by 2050 or sooner has been established
- Individual investment teams have, as part of their existing philosophy and process, committed to engage with their strategy's highest carbon-emissions contributors with the aim of improving investment outcomes

We expect this initial interim commitment to grow, and we will share our progress over time. In line with our accountability-driven approach, we initially restrict our asset-level commitment to strategies invested predominantly in equities, investment-grade bonds, and high-yield bonds, as emissions data and accepted decarbonization methodologies are currently only available for these asset classes. We plan to expand the scope of addressable asset types over time as better data and methodologies emerge. We hope to add strategies focused on sovereigns, derivatives, short exposures, and private securities.

For details on our implementation approach, please see 'Our commitment to net zero' white paper, available on our website.

Section 3: Engagement philosophy, approach, and examples

We rely on engagement to assess companies' credibility on climate-risk management. Because most emissions data we need to build analytical tools is backward-looking, incomplete, or estimate-based, engagement helps us evaluate forward-looking investment risks and opportunities.

DRIVING OUTCOMES THROUGH ACTIVE OWNERSHIP

Each year, more clients ask us for help in pursuing their climate objectives. Company engagement enables us to do that. In our engagements, we typically start by assessing companies' understanding of climate physical and transition risks. Many companies do not yet disclose climate-relevant information, so we may begin by encouraging them to measure and disclose their Scopes 1, 2, and 3 emissions as well as the physical locations of property, plant, and equipment.

Over time, regular engagement allows us to influence corporate behavior in ways we believe can unlock value, by helping companies adopt best practices for climate-risk management transparency, for example, and setting science-based emissions-reduction targets. As more companies improve, we intend to intensify our focus on the credibility of climate-risk management, including incentives alignment and capital-allocation decisions. We actively track and measure engagements to monitor outcomes, assess effectiveness, and identify candidates for escalation. Our portfolio teams are familiar with the escalation tools at their disposal, including proxy voting and public forums via our external partnerships.

2021 ENGAGEMENT PRIORITIES

Financial disclosures: We have been engaging with portfolio companies to encourage the adoption of the TCFD for several years. In 2021, we began to see management teams respond. TCFD-aligned disclosure is evidence of a climate-risk-management process. TCFD-aligned disclosures help us assess the rigor of the company's climate strategy and focus on areas of improvement.

Transition-risk mitigation: We believe mitigation requires setting and meeting challenging, time-bound SBTs to reduce emissions. Demonstration of progress signals a company's willingness and ability to mitigate its transition risks over time. We monitor whether and how companies link targets to capital allocation decisions and executive compensation. This is an area of increasing focus for investment teams with net-zero-aligned client accounts.

Physical-risk adaptation: We find most companies focus less on identifying and addressing physical climate risks than on mitigating transition risks. We would like more companies to disclose their expected financial impacts of physical risks under multiple climate scenarios, and for management teams to proactively adapt to these risks. In 2019, we published our Physical Risks of Climate Change (P-ROCC) framework designed to help companies identify, analyze, and disclose their exposure to physical climate risks. In 2021, we expanded the framework with P-ROCC 2.0, which calls for companies to disclose physical location data to help investors better assess their risks and discuss building operational resilience.

Leading companies: We continue to engage with companies we believe are leading their peers on climate-risk management or opportunity alignment. Climate leaders can often provide real-world examples of new or evolving best practices in a particular industry or region. Insights we learn in these engagements help us improve our feedback to other companies.

ENGAGEMENT EXAMPLES

Climate change has long been an engagement priority for Wellington. In 2021, we conducted over 17,500 meetings with 4,500 public-market issuers in 97 countries. We voted on approximately 67,000 resolutions at more than 6,600 shareholder meetings. In 2021, climate transition and/or physical risks were a substantive agenda item in meetings and proxy votes, and as such we had approximately 1,400 climate-related meetings.

We chose the following examples to demonstrate multiple forms of discussions on client transition and/or physical risks, broad involvement from investors across asset classes and regions, and a range of engagement outcomes. They are a subset of the firm's list of company engagements on climate change during 2021, shown in Appendix 3. We selected examples based on their ability to demonstrate our engagement and voting policies in action in accordance with the recommendations of the TCFD. Many of our engagements did not result in meaningful or measurable outcomes. We continue to develop our engagement tracking tools to enhance research and integration, as well as client reporting.

AKER

- **Primary topic**: Impact of carbon pricing on business-mix evolution
- **Method**: Engagement with management
- **Detail**: Aker engages in the exploration, development, and production of oil and gas on the Norwegian continental shelf. We engaged on environmental topics, climate transition, and target-setting. We believe economic and regulatory incentives are motivating management to accelerate carbon-emissions reduction. While the carbon intensity of Aker's production is among the lowest in its industry, the company has goals to reduce it further. Management's long-term planning has benefited from high carbon prices, and the company is increasing its carbon-price assumptions for capital allocation decisions. The bulk of Aker's carbon-efficiency gains stem from improved energy efficiency and electrification of offshore platforms.
- **Outcome**: Overall, our engagement suggested a positive outlook for management of environmental risks. Prior to engagement, we believed Aker had stronger management of ESG risks than its exploration and production peers. Our engagement solidified this conviction. We will continue to engage with Aker to understand the methodology it uses to assess alignment with the Norwegian government's climate targets. Additionally, we will seek to understand how the company plans to build resilience to material risks of flooding and subsidence (sinking ground level) to its platforms in the North Sea, identified as risks in Aker disclosures and corroborated by our Climate Research Team.

AMERICAN TOWER

Primary topic: Decarbonization target-setting

Method: Engagement with investor relations and sustainability teams

Detail: American Tower (AMT), one of the largest global REITs, owns, operates, and develops a global portfolio of communications real estate. We engaged with members of AMT's investor relations and sustainability teams to discuss the company's carbon-footprint measurement and reduction targets. AMT acknowledges the increasing importance to investors and customers of emissions reduction. This is evident in its work in emerging markets and candor on how greater energy efficiency can help its customers advance their goals and lower costs.

Africa and Asia account for more than 90% of the company's emissions. To date, AMT has completed more than half of its 10-year goal, focused on those regions. We encouraged the company to be thoughtful about communicating how its on-the-ground efforts translate into overall target-setting as a competitive advantage for attracting and retaining customers and appeasing stakeholders. At the time of our engagement, the company was more than two years into its validation process with the Science Based Targets initiative (SBTi), following its initial commitment. During this period, AMT had measured and reported its Scope 3 emissions — a significant component of its overall footprint — given that it does not directly operate all of its sites. AMT was receptive to and inquisitive about our process for assessing transition-risk management and target robustness.

Outcome: AMT adopted SBTs in line with a well-below-2°C scenario in late 2021. Specifically, it has committed to reducing Scopes 1, 2, and 3 emissions by at least 40% by 2035. This effort is particularly notable given the use of diesel fuel in emerging markets lacking grid access. AMT has effectively committed to replacing diesel with renewables and storage systems in its emerging market operations.

CHEVRON

Primary topic: Decarbonization and net zero

This meeting was a continuation of our previous engagements with the company on its physical-risk exposure and approach to the low-carbon transition. We were seeking more information on how the company plans to respond to a shareholder proposal to reduce emissions and what strategic options are available.

Method: Engagement with board

Detail: We have had ongoing engagements with Chevron on climate change. We engaged with the chair and CEO to discuss management's response to a 2021 shareholder proposal that received 61% support requesting the company reduce its Scope 3 emissions. This was a follow-up engagement to a meeting earlier in the year. We reiterated our message that committing to net zero by 2050 aligns with the company's long-term strategy for the energy transition, so making a commitment would bolster its investment case. We had supported the shareholder proposal, and during our engagement we also discussed the company's strategic options.

Outcome: We observed that Chevron is responding to investor concerns about climate. In October 2021, the company issued an updated climate-resilience report, detailing its adoption of a 2050 net-zero aspiration for upstream Scopes 1 and 2 emissions. The report also detailed how Chevron was incorporating Scope 3 emissions into its GHG emissions targets by establishing a portfolio carbon intensity target, inclusive of Scopes 1 and 2, as well as Scope 3 emissions from the use of its products. While these actions show progress, the company will remain a priority for engagement.

COMPAGNIE DES ALPES

Primary topic: Preparing for warmer, shorter winters

Method: Call with management

Detail: For this operator of French ski resorts and theme parks, engagement with a focus on the environment is a critical part of assessing its investment case. We engaged to discuss the company's exposure to physical risks and how it plans to build resilience. During our call, we discussed at length its past investments in artificial snow production, as well as its strategies to reduce its carbon footprint. We gained conviction in the long-term viability of Compagnie de Alpes' business model by leveraging our climate research. Specifically, we assessed historical and changing projected patterns of rising temperatures and mapped the potential impacts onto the company's resort portfolio across France.

Outcome: We believe there is an opportunity to reengage, share this information, and deepen our strategic climate discussion with the company.

HCA HEALTHCARE

Primary topic: Capacity management amid increasing hurricane frequency

Method: Engagement with management

Detail: HCA Healthcare is a health care services company operating general- and acute-care hospitals and outpatient facilities in the US and UK. With 50% of the company's hospitals located in Florida and Texas (accounting for 48% of revenue), HCA faces significant physical climate risk. In 2018, the company suffered lost revenue and incurred expenses totaling US$31 million following Hurricane Michael. During our engagement, we encouraged the company to disclose details about its hurricane-related risk-mitigation process and be more precise about specific geographical risks, consistent with our P-ROCC 2.0 guidance. HCA has a cross-functional emergency preparedness team and insurance policies to cover business disruption caused by extreme events.

Outcome: We concluded the company could more proactively address and communicate its resiliency planning and preparedness, an issue that had negatively influenced our ESG rating. We will continue to evaluate this climate risk, weighing it against strong population growth and positive demographic trends in Florida and Texas.

ENGAGING ON CLIMATE TRANSITION PLANNING

Our approach to engagement on alignment with the low-carbon transition acknowledges that companies are at distinct stages of their transition journeys depending upon their industry and region. While the long-term goal is the same, we seek to structure our engagements to meet companies where they are. Our aim is to improve on their current position and converge to industry-specific best practice over time. Therefore, we tailor engagement questions to each company. Companies' current alignment varies across a continuum: transparency, ambition, and credibility.

TRANSPARENCY

We expect companies to have credible transition plans communicated using the recommendations of the TCFD. We view disclosure of Scopes 1 and 2 emissions as a minimum expectation where measurement practices are well-defined and attainable. We encourage all companies to disclose Scopes 1, 2, and 3 emissions. While we recognize the challenges associated with collecting Scope 3 emissions data, this disclosure is necessary for us to fully understand the transition risks applicable to an issuer. Disclosure of both overall Scope 3 categories — upstream and downstream emissions — with context and granularity from companies about the most significant Scope 3 sources enhances our ability to evaluate investment risks and opportunities.

AMBITION

As an outcome of enterprise risk management and strategic planning to reduce the potential financial impacts of climate change, we encourage companies to set a credible, science-based decarbonization glidepath. When evaluating interim and long-term targets, we consider two primary elements:

- Comprehensiveness: Does the target comprise all categories of material emissions?
- Rigor: Is the target consistent with the ambition to achieve net-zero emissions by 2050 or sooner?

We consider it to be best practice for companies to pursue validation from the SBTi.

CREDIBILITY

Once a target is set, we continue to monitor companies' implementation, seeking additional data to reinforce strategic alignment of a decarbonization strategy with corporate strategy. These data points include capital allocation, management incentives, and lobbying practices. We believe in prioritizing emissions abatement within the value chain. When a company uses offsets to neutralize residual emissions, the offsets should be high in quality and remove or reduce GHG emissions in real, additional, and permanent ways. They should also have minimal negative social or environmental impacts ("do no significant harm").

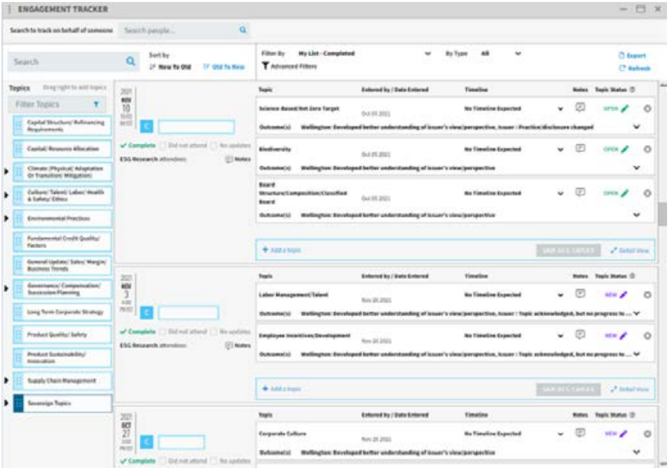
TRACKING ENGAGEMENT OUTCOMES

We actively track progress toward our engagement objectives and monitor for changes in practices or disclosures. Our engagement tracker (**Figure 5**) is a shared tool investors across our firm can use to record and collaborate on engagement topics. It facilitates tracking of engagements for escalation purposes. Once booked, all investors are invited to engagements through a central calendar. Bringing asset-class specialists together gives company managements a window on the various investor perspectives, such as balance-sheet leverage, cost of capital, and material ESG issues. We believe this transparency helps ensure the representation and consideration of varied client interests on material engagement matters.

We track all engagements to ensure we have the ability to continue conversations over multiple years and see progression and consistency in our messaging. In 2019, we enhanced our engagement-tracking capabilities by creating a tool available to all investment teams. Teams can log engagement details, including topics, outcomes, and timelines for reviewing progress in the future. In 2021, we integrated our ESG and fundamental engagement-tracking tools, enabling investment teams to easily incorporate ESG engagement information into their investment processes. By moving to one tracking system, we have been able to better evaluate and monitor our climate engagement activities in the context of our overall engagement. This system will also help inform future engagement and proxy voting plans. Our goal is to become more effective in our dialogue with companies over time.

Figure 5

Proprietary engagement tracker



For illustrative purposes only. Source: Wellington Management

ESCALATION AND PROXY VOTING

As a complement to engagement, we can use proxy voting as part of our escalation framework for managing climate-related risks. Voting provides leverage for discussions with management and opportunities to directly influence corporate policy. We evaluate shareholder proposals related to environmental and social issues on a case-by-case basis and expect portfolio companies to comply with applicable laws and regulations regarding environmental and social standards. We consider the spirit of a proposal — not just the letter — and typically support proposals addressing material issues even when management has been responsive to our engagement on the issue. In this way, we seek to align our voting with our engagement activities.

In 2021, we supported 42% of environmental shareholder proposals globally, and 54% of US proposals.

We generally support shareholder proposals asking for improved disclosure on climate-risk management and we expect to support those requesting alignment of business strategies with the Paris Agreement or similar language. We also generally support proposals asking for board oversight of political contributions and lobbying activities or those asking for improved disclosures where material inconsistencies in reporting and strategy may exist, especially as it relates to climate strategy. If our views differ from any specific suggestions in the proposals, we will provide clarification via direct engagement. Additionally, as part of the firm's 2021 annual voting policy review, we decided to impose more accountability for directors on lack of progress with Scopes 1 and 2 emissions disclosure. Effective in 2022, we will vote against the chairs of companies in the MSCI World Index and issuers covered by the Transition Pathway Initiative that fail to disclose Scopes 1 and 2 emissions data.

COLLABORATIVE ENGAGEMENT AND EXTERNAL PARTNERSHIPS

COLLABORATIVE ENGAGEMENT

As a member of Climate Action 100+, we are currently involved in six engagements in which we believe we can meaningfully contribute knowledge. The engagements are with Phillips 66, International Paper, American Electric Power Company, ConocoPhillips, Kinder Morgan, and Standard Chartered. ESG analysts serve as primary representatives, and portfolio managers with substantial holdings add their voices to these ongoing engagements.

Collaborative engagement complements our ongoing private dialogue with companies. Across both private and collective forums, we continue to support the three main asks of Climate Action 100+ board accountability and oversight, TCFD-aligned disclosure, and a decarbonization target. Wellington is also a member of the Investor Forum, a UK-based collaborative network. Through its collective engagement framework, the Forum consults with members to hear their concerns, identify key issues, and develop constructive solutions.

Notable updates from our Climate Action 100+ collaborative engagements in 2021 include:

INTERNATIONAL PAPER

- Encouraged better transparency around lobbying tied to climate-related topics, including positions with trade groups such as the American Forest and Paper Association

KINDER MORGAN

- Met with members of Kinder's Health, Safety & Environment (HSE) team to discuss progress on net zero, reporting on Scope 3, and the Climate Action 100+ benchmark
- Met with the company's president on a range of topics, including expanding carbon disclosure on a utility gas supply chain, initiatives with its energy transition group (including landfill gas and renewable diesel opportunities), the IEA's Net Zero Pathway report, methane management practices, and consideration of joining the Oil & Gas Methane Partnership (OGMP)

STANDARD CHARTERED

- Provided feedback on net-zero disclosure/roadmap and communication with investors through Investor Forum ahead of 2022 annual general meeting
- Separately from Investor Forum, we engaged with the remuneration committee on how companies link management compensation to sustainable-finance goals

EXTERNAL PARTNERSHIPS

Along with our extensive internal research on sustainable investing, we partner with leading organizations to educate ourselves and provide leadership on asset management perspectives. We communicate with experts, from climate scientists to academics, on a range of topics with the intention of deepening sustainability insights.

Partnership	Role	Date
Responsible investment		
UN PRI	Signatory	2012
PRI Statement for Credit Ratings	Signatory	2016
Focusing Capital on the Long Term (FCLTGlobal)	Member	2017
Council of Institutional Investors	Member	2018
Investor Forum (UK)	Member	2019
PRI's Venture Capital Network	Member	2021
Climate change		
Statement of Support for TCFD	Supporter	2017
Transition Pathway Initiative	Supporter	2018
CDP (formerly Carbon Disclosure Project)	Member	2019
Ceres Investor Network on Climate Risk and Sustainability	Member	2019
Climate Action 100+	Member	2020
IIGCC	Member	2020
Stewardship		
UK Stewardship Code	Signatory	2010
Japan Stewardship Code	Signatory	2014
Hong Kong Principles of Responsible Ownership	Signatory	2016
Investor Stewardship Group's Governance and Stewardship Principles	Founding member	2017
FSC Standard 23 Australia	Signatory	2018
Corporate governance		
Asian Corporate Governance Association (ACGA)	Member	2007
International Corporate Governance Network (ICGN)	Member	2012
Impact investing and ESG		
UN SDGs	Supporter	2015
GRESB (formerly Global Real Estate Sustainability Benchmark)	Member	2015
GIIN	Member	2016
Impact Management Project	Practitioner	2019
Sustainability Accounting Standards Board (SASB) Alliance	Member	2021
ESG Data Convergence Project	Signatory	2021
Institutional Limited Partners Association—Diversity in Action	Supporter	2021

IDENTIFYING CLIMATE-RELATED INVESTMENT OPPORTUNITIES

Our research partnerships with Woodwell and the MIT Joint Program give us high conviction that climate change is a secular theme requiring significant capital investment. In our view, the market continues to underestimate the spending needed to help society adapt to and mitigate climate change. As physical risks intensify, the need for adaptation-related spending will almost certainly increase. And, because physical risks accelerate the transition to a low-carbon global economy, the need for solutions to help companies mitigate climate-related transition risks should grow commensurately.

Our work has uncovered investment opportunities across public and private markets in companies whose products and services facilitate the low-carbon transition and help society adapt to physical climate change (**Figure 6**). In monitoring the constantly evolving market for climate solutions, we continue to identify near- and longer-term risks and opportunities across sectors, geographies, and asset classes.

Figure 6

Broad investable universe of climate solutions

	ENERGY	BUILDINGS/ INFRASTRUCTURE	TRANSPORTATION	INDUSTRIAL	FOOD/ AGRICULTURE
MITIGATION	Renewables-facilitation technology Grid optimization Storage software Financing models	HVAC innovation Building optimization Design/simulation software Water efficiency	EV and fleet software Shared/micro mobility Connected transportation Logistics optimization	Intelligent manufacturing Process optimization Supply-chain optimization Recycling technology	Protein alternatives Biotechnology Farm-to-consumer Waste reduction Smart-nutrition devices
ADAPTATION	Utility resilience Backup power Microgrids	Climate-risk analytics and insurance Emergency-response technology Weatherization Water security	Logistics resilience Infrastructure planning and monitoring	Industrial cybersecurity Supply-chain resilience Field service management	Controlled-environment agriculture Farm management software Food safety and traceability

Source: Wellington Management. There is no assurance that any climate investing strategy and techniques employed will be successful and investments can lose value. Please refer to the Important disclosures at the end of this report.

Section 4: Metrics and targets

As we encourage investee companies to improve climate disclosures, we understand the importance of leading by example. In addition to deepening our understanding of the effort required (many companies cite measurement as their greatest challenge in adhering to the TCFD framework), we can share with peers and investee companies how we sought to address our own measurement challenges. A core component of the NZAM commitment is the belief that transparency fosters accountability and provides opportunities to demonstrate differentiation.

Our community-of-investment-boutiques structure means we do not have a singular investment view. Our investment teams each function as entrepreneurial entities. For the same reason, we do not manage firmwide climate-risk exposure. This responsibility falls to individual portfolio management teams, according to their particular strategy's mandate. All teams have access to the same metrics, however, allowing us to assess aggregate equity and corporate credit climate-risk exposure. We recognize that there is no single, comprehensive metric through which to assess our portfolios' exposure to climate risks. As such, we leverage a mosaic of metrics to conduct climate-risk analysis and assess net-zero alignment at the portfolio level. These metrics include current and projected weighted average carbon intensity (WACI), financed emissions, exposure to carbon-related assets, implied temperature rise (ITR), and climate value-at-risk (CVaR).[4]

We are able to calculate GHG footprints for our investment strategies invested in corporate and sovereign securities, as relevant data for these asset classes is most widely available. Our investment teams can use this information to assess a portfolio's overall emissions footprint — including the distribution of carbon intensity across sectors and regions — and to identify top emissions contributors. This information is available to clients as a standard component of our ESG reporting as well.

4 The analysis below is based upon aggregate gross long equity and corporate credit exposure of nearly US$750 billion and US$300 billion, respectively, for total corporate exposure of more than US$1 trillion (representing about 75% of total assets under management). Analysis is based on holdings and carbon data available as of 31 December 2021, unless otherwise noted. Scopes 1 and 2 emissions are based upon a combination of company-disclosed and estimated data to supplement data gaps, while Scope 3 emissions are based upon a fully estimated dataset; both are sourced from MSCI. These figures have not been independently assured or audited.

MEASURING CURRENT EXPOSURE ACROSS OUR EQUITY AND CORPORATE CREDIT PORTFOLIOS

Note: This section focuses on our aggregate exposure. Appendix 2 shows the same metrics for a sample of our equity and corporate fixed income strategies, as we disclose them to clients.

WEIGHTED AVERAGE CARBON INTENSITY

Because we believe that, in general, energy-efficient companies manage transition risks better than their less-efficient peers, WACI is our preferred portfolio-emissions metric. WACI measures a portfolio's exposure to carbon-intensive companies, using corporate revenue as a proxy for size. WACI also enables us to compare companies across industries. When conducting peer analysis, our industry research analysts may use a more specific intensity-based metric, such as kilowatt hours, for utilities.

For the purposes of this report and standard aggregate portfolio reporting, we use revenue as an approximation for output, in line with the TCFD's Supplemental Guidance for Asset Managers. Other recommended portfolio metrics, such as Carbon Footprint and Total Carbon Emissions, use an ownership approach.

Because WACI changes only with annual reported emissions or revenue, the metric is not sensitive to share-price movements or other changes in capital structure and output. WACI is accessible and easy to communicate to clients. And like returns-based attribution analysis, WACI enables us to decompose a portfolio to clearly illustrate how both sector allocation and security selection influence a portfolio's intensity.

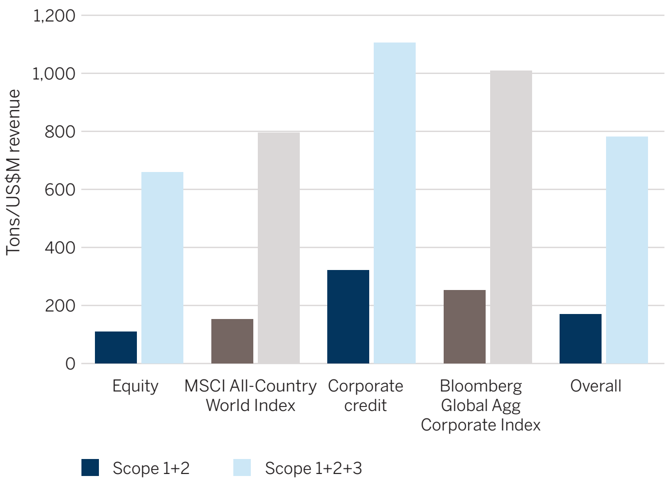
Figure 7 shows the WACI of our equity and corporate credit exposures, relative to a representative global universe,[5] as of year-end 2021. Based on an attribution analysis, Wellington's equity WACI (Scopes 1 and 2) compares favorably to a global universe; we attribute this to stock selection within the two sectors representing the largest contribution to the footprint: utilities and materials.

While our equity exposure has a higher allocation to utilities than the universe, our portfolios, in aggregate, are weighted toward more carbon-efficient utilities — those with less-carbon-intensive energy-generation mixes than global peers, on average. Our corporate-credit assets have a higher WACI than the index because of less favorable security selection in utilities, the sector that represents the largest contribution to the footprint. In addition, the representative index is comprised of investment-grade issuance, while our corporate-credit exposure includes investment-grade and high-yield issuance. For reference, the WACI for a global high-yield universe is more than 50% higher than that of a global investment-grade universe.[6]

While we undertake the same attribution analysis for the WACI for Scopes 1, 2, and 3, we proceed with caution, as we expect the estimated dataset to be merely directionally accurate at this time. The analysis indicates underweights to energy and materials, as well as an overweight to health care, which contribute to a favorable sector-allocation profile. We will seek to analyze the stock-selection effect as Scope 3 disclosure improves, in order to provide more differentiation within industries. We expect to be able to connect stock-selection information to our fundamental analysts' recommendations about the drivers of long-term value. For example, within the automotive industry, we should see lower Scope 3 downstream emissions from manufacturers offering a larger share of electric vehicles, relative to peers.

Figure 7

WACI of Wellington's equity and credit exposures



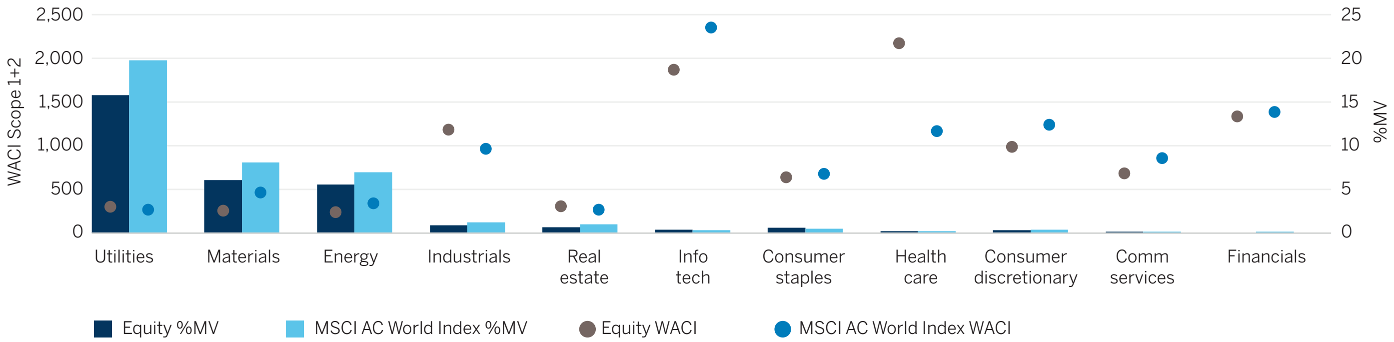
Data as of 31 December 2021. Sources: MSCI, Bloomberg, and Wellington Management.

5 MSCI All Country World Index has been used to represent the global equity universe, and the Bloomberg Global Agg Corporate Index has been used to represent global investment grade, fixed-rate corporate debt. Comparisons are for informational purposes only; not all assets aggregated as Wellington's equity and corporate credit exposure are managed against these indices.

6 Wellington analysis as of 31 December 2021, based on a comparison of Bloomberg Global Agg Corporate Index and Bloomberg Global High Yield Index.

Figure 8

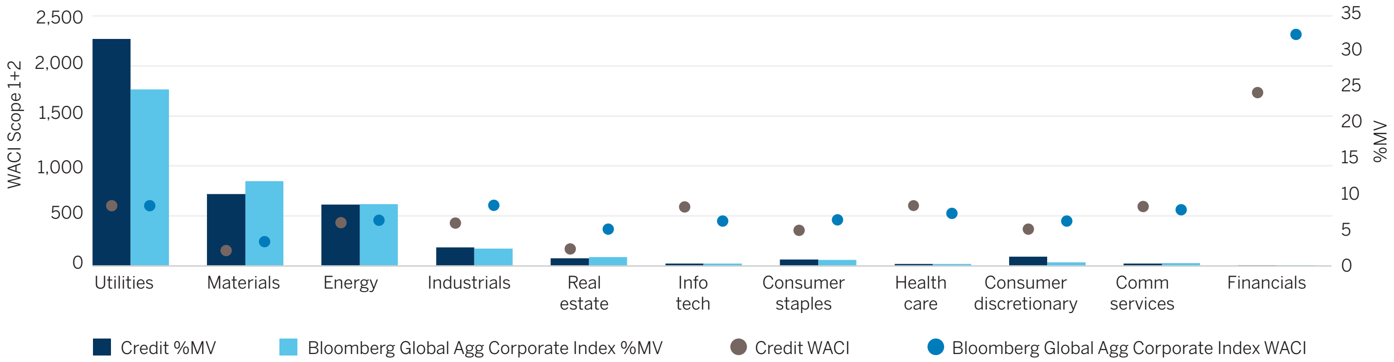
Sector-level breakdown of Wellington's WACI for equity exposure



Data as of 31 December 2021. Sources: MSCI and Wellington Management.

Figure 9

Sector-level breakdown of Wellington's WACI for corporate-credit exposure



Data as of 31 December 2021. Sources: MSCI, Bloomberg, and Wellington Management.

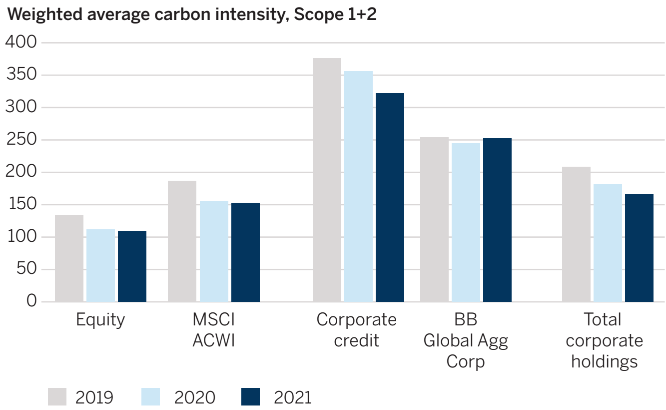
A sector-level analysis comparing our equity and corporate-credit exposure to representative universes can provide further insight. **Figure 8** demonstrates that Wellington's equity investments tend to have a lower sector-level WACI than the index — particularly in intensive sectors like utilities, materials, and energy — and that Wellington is modestly underweight these sectors. On the other hand, the overweight to less-intensive sectors like health care also contributes to a lower WACI.

Figure 9 shows that the WACI for Wellington's corporate-credit investments is driven largely by the higher sector-level WACI in utilities compared to the index. Note that WACI does not account for the evolving generation mix of utilities over time. Contrary to the trend in equities, the underweight to less-intensive sectors like real estate and financials also contributes to a higher WACI.

The WACI of our equity and corporate-credit portfolios have decreased over the last two years by 19% and 14%, respectively (**Figure 10**). The drop in equity WACI is consistent with the 18% decrease in the global equity index, suggesting that some of the improvement is due to natural decarbonization of the broader universe. The decrease in corporate fixed income WACI stands out against the global credit index, where no progress was observed over the period. Security selection in carbon-intensive sectors has improved, however, narrowing the gap with the index. For example, the WACI for utilities, which accounts for two-thirds of our overall fixed income WACI, has decreased by 17% since 2019. Currently, green bonds are attributed to an issuer as if they were general issuance bonds. This limitation likely results in a consistently overestimated intensity measure for both a portfolio and the index.

Figure 10

Change in WACI of Wellington's portfolios over time



Data as of 31 December 2021. Sources: MSCI, Bloomberg, Wellington Management.

CARBON FOOTPRINT: EMISSIONS ASSOCIATED WITH ASSETS UNDER MANAGEMENT[7]

As a complement to WACI (which highlights the emissions efficiency of a portfolio), investors also want to understand how their portfolios contribute to climate change. Carbon footprinting is how financial institutions allocate emissions to "Scope 3 Category 15, Investments," per the Partnership for Carbon Accounting Financials (PCAF) Standard. To measure this, we use the ownership approach, which allocates each company's emissions to its investors based on an investor's share (amount invested) of the company's enterprise value, including cash (EVIC). If an investor owns 1% of a company's EVIC, then the investor's strategy is responsible for 1% of the company's emissions.[8] The financed emissions metric per US$ million invested is a normalized version of the same figure, where total financed emissions are divided by asset class AUM. This metric can facilitate comparison among various-sized managers.

Figure 11 shows that the equity and credit assets Wellington manages on behalf of our clients contribute fewer emissions than global indices. This is true across Scopes 1 and 2 and Scopes 1, 2, and 3 emissions. On average, every US$1 million AUM in a Wellington equity strategy contributes 33.0 tons of carbon emissions; the same US$1 million passively invested in the MSCI All Country World Index contributes 52.9 tons. In our credit portfolios, every US$1 million invested contributes 55.8 tons, while the same amount invested in the Bloomberg Global Agg Corporate Index contributes 67.6 tons.

Figure 11

Financed emissions from assets under management

	Scopes 1 and 2		Scopes 1, 2, and 3	
Equity	**WMC***	**MSCI ACWI**	**WMC**	**MSCI ACWI**
Financed emissions (million tCO2e)	24.1	39.1	175.6	232.0
Financed emissions per US$ million invested (tCO2e)	33.0	52.9	235.1	310.0
Credit	**WMC**	**BB Glb Agg Corp**	**WMC**	**BB Glb Agg Corp**
Financed emissions (million tCO2e)	14.5	17.6	95.8	108.3
Financed emissions per US$ million invested (tCO2e)	55.8	67.6	370.2	420.6

*WMC used here for Wellington Management Company. Data as of 31 December 2021. Sources: MSCI, Bloomberg, and Wellington Management.

Since financial institutions must rely upon estimates of carbon emissions to calculate portfolio emissions, the PCAF Standard suggests disclosure of a data-quality score on a scale from 1 (highest quality, verified disclosure) to 5 (lowest quality, estimation based on asset turnover ratios). Based on the MSCI dataset leveraged for this analysis, we conclude the data-quality score for both equity and corporate credit asset classes is between 2 and 3. The majority of market value within the global universe is directly reported but unverified, while the remainder is estimated by MSCI using a combination of physical activity and revenue factors. We expect the data-quality score to improve as regulators increasingly require climate disclosure and as investors pressure companies to voluntarily provide this information.

In the process of adopting the PCAF Standard for financed emissions, we have encountered a few challenges requiring further consideration. While we continue to report this calculation per clients' requests, we prefer not to use it as the basis for portfolio construction or guideline development. We leverage a data field inclusive of Market capitalization at fiscal year-end date + Preferred stock + Minority interest + Total debt (short- and long-term), and cash/cash equivalents. This places equal weight on all issuance regardless of duration and presumes shorter-term debt continually rolls over. PCAF acknowledges that "specific elements of EVIC might not be readily available because data providers are still working to align their data with this definition," so this metric should become more mainstream over time.

Since we must make availability of EVIC a condition of our calculation, this can sometimes lead to small reductions in coverage (and rescaling of covered securities) unrelated to carbon-emissions availability. We are also cognizant of counterintuitive relationships when analyzing changes in financed emissions metrics over time. For example, financed emissions attributable to one investment can change absent a change in absolute emissions or intensity; instead, the capital structure as represented by EVIC may cause the shift. We are following developments in accounting for market volatility to establish a level of standardization for this EVIC normalization and may restate these figures once adjustments to the standard calculation are incorporated.

7 This analysis is consistent with the Public Carbon Accounting Financials (PCAF) Global GHG Accounting & Reporting Standard for the Financial Industry (PCAF Standard), based on holdings as of 31 December 2021 and carbon emissions and ownership data available from MSCI as of 31 December 2021. Given the lag in carbon emissions disclosure and estimation, the data available at this date is a combination of 2018, 2019, and 2020 fiscal years.

8 Limited data at the issuer level remains the main challenge in calculating financed emissions; however, issuer coverage has been improved by use of estimated data, for Scope 3 emissions in particular, in order to identify carbon-intensive market segments invested as well as to compare potential for progress across them.

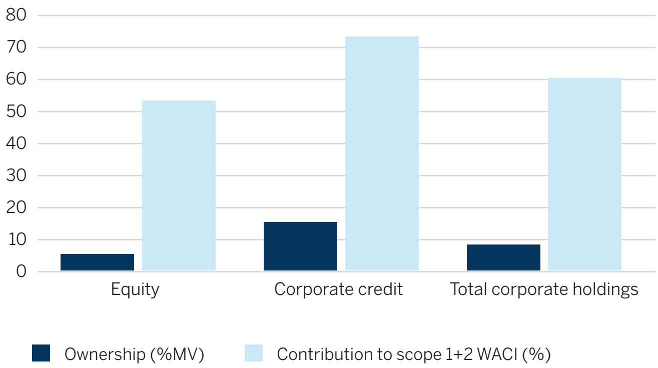
EXPOSURE TO CARBON-INTENSIVE SECTORS

The TCFD's financial institutes define carbon-related assets as "those assets tied to the energy and utilities sectors under the Global Industry Classification Standard, excluding water utilities and independent power and renewable electricity producer industries." Consistent with this definition, **Figure 12** shows carbon-intensive sectors represent 8% of invested corporate assets (dollars invested as a proportion of AUM in corporate asset classes) and contribute disproportionately to the overall Scopes 1 and 2 WACI (60% contribution). The electric utilities industry is the largest contributor to the WACI of both asset classes.

Because many companies embrace their role in the energy transition, we find this to be a blunt instrument with which to measure exposure. Nonetheless, some companies also have high Scope 1 emissions, making them likely targets of carbon-pricing schemes in their respective regions, underscoring the materiality of transition risk and importance of assessing transition plans in these industries. Within these carbon-intensive industries, we seek to pair Scopes 1 and 2 intensity with a more forward-looking approach integrating companies' emissions-reduction targets. Within carbon-intensive industries, we leverage supplemental disclosures about capital expenditure plans to gain conviction in companies' ability to achieve stated targets.

Figure 12

Wellington's exposure to carbon-intensive sectors (%)



Data as of 31 December 2021. Source: Wellington Management.

INTEGRATING FORWARD-LOOKING METRICS

SCIENCE-BASED EMISSIONS-REDUCTION TARGETS

In addition to the top-down measurement from carbon-footprint metrics, investment teams also use a bottom-up measurement to highlight the proportion of underlying holdings with robust decarbonization targets. We use the public dataset provided by the SBTi as a proxy for the universe of robust targets, with the understanding that companies may not have pursued target validation from SBTi. We encourage companies to establish emissions-reduction targets consistent with guidance from SBTi, which helps them to develop net-zero pathways and transition plans aligned with a 1.5°C future. We expect to continue to use engagement to encourage adoption of this best-practice standard.

When measuring portfolios, we assess exposure in terms of a holding's market value and its contribution to WACI. We want to understand whether or not carbon-intensive companies — with material transition risk — are adopting targets. The greater proportion of a portfolio's carbon footprint covered by targets, the faster the pace of organic portfolio decarbonization we can expect.

SBTi has two qualification statuses: "Targets set" and "Committed." Targets set indicates that a company's target has been validated as comprehensive, robust, and aligned with approved methodologies. Committed indicates that a company is in the process of setting targets and seeking review from SBTi. Because the target-setting and approval process takes time, we expect companies to move from Committed to Targets set within two years of a commitment date.

Figure 13

Portfolio companies with SBTs and contribution to WACI vs index

Equity				
	Exposure to SBTs (%MV of Wellington equity)	**Contribution to Wellington equity WACI (%)**	**Exposure to SBTs (%MV of MSCI ACWI)**	**Contribution to MSCI ACWI WACI (%)**
Targets set	26.6%	14.9%	26.9%	16.4%
Committed	10.8%	8.1%	12.8%	8.5%
Total	**37.4%**	**23.0%**	**39.7%**	**24.9%**
Credit				
	Exposure to SBTs (%MV of Wellington corporate credit)	**Contribution to Wellington corporate credit WACI (%)**	**Exposure to SBTs (%MV of BB Glb Agg Corp)**	**Contribution to BB Glb Agg Corp WACI (%)**
Targets set	30.1%	13.6%	23.5%	13.9%
Committed	10.4%	3.5%	12.8%	5.6%
Total	**40.4%**	**17.0%**	**36.3%**	**19.4%**

Data as of 31 December 2021. Sources: MSCI, Bloomberg, SBTi, Wellington Management.

In **Figure 13**, we mimic this bottom-up approach to portfolio-level analysis for Wellington's overall equity and corporate-credit assets for illustrative purposes, which is distinct from our AUM commitment to the NZAM. Across Wellington's equity exposure, exposure to companies with SBTs or commitments to SBT is approximately 37%, roughly in line with global equity-market coverage for both market value (MV) and carbon footprint. Our corporate-credit exposure to companies with SBTs is about 40%, slightly higher than global corporate-credit markets coverage for market value and slightly lower for carbon footprint.[9] We applaud the pace of company adoption of SBTs over the last two years. In equity markets, the 39.7% covered by a target or commitment at year-end 2021 is a nearly threefold increase from 13.9% of MSCI ACWI by market value at year-end 2019. Credit markets have shown similar growth: 36.3% covered by a target or commitment at year-end 2021 is a significant increase from 12.9% in 2019.

PROJECTED WACI

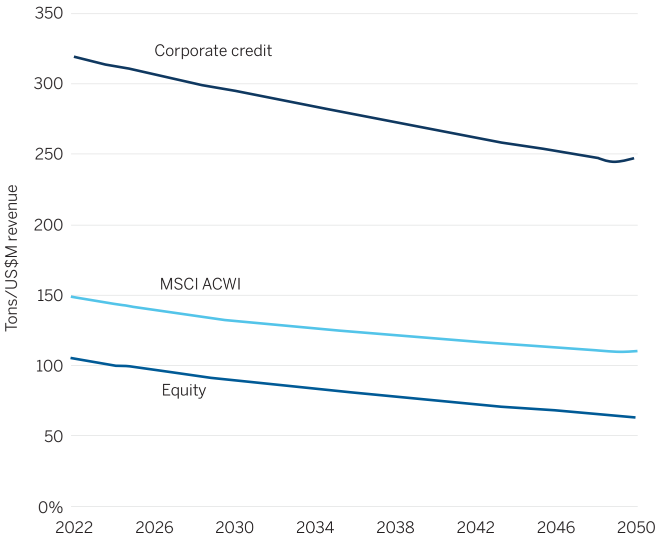
Typically, WACI is backward looking, and because companies must report this data with a time lag, WACI does not reflect progress toward targets. To encourage target-setting, we use projected emissions data to connect companies' decarbonization plans with anticipated changes in portfolio WACI. As **Figure 14** shows, we project WACI in 2030 to demonstrate the amount of natural or bottom-up decarbonization we expect to occur as our portfolio companies execute against their stated targets. Projections are based on current emissions and reduction targets, if available. (This information incorporates targets of any rigor, not only those targets validated by SBTi.)

We assume companies with targets will achieve them. We assume companies without a current target will not set one, thus flatlining their carbon intensity through 2030 (with emissions and revenue growing at the same rate). The latter assumption is a conservative estimate for our aggregate WACI, as we expect to increase portfolio exposure to companies with reduction plans through engagement.

Based on current holdings and their targets, equity exposure is on track to reduce WACI by 30% by 2030, relative to a 2019 baseline. As we engage to encourage more carbon-intensive companies to adopt targets, the slope of this projected WACI line should continue to steepen.

Figure 14

Projected WACI (Scopes 1 and 2) of equity and corporate credit assets under management



Data as of 31 December 2021. Source: Wellington Management. Data subject to numerous assumptions and may change based on factors that include, but are not limited to, trades, additional company targets, and company execution against targets.

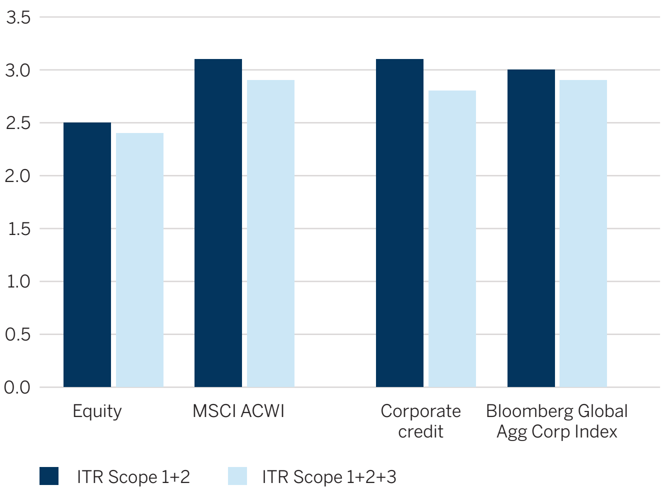
9 Equities proxied by the MSCI All Country World Index, credit proxied by the Bloomberg Global Agg Corporate Index.

IMPLIED TEMPERATURE RISE

Another increasingly popular means of measuring portfolio alignment is implied temperature rise (ITR). Expressed in Celsius, ITR aims to quantify the degree to which a company's activities against sectoral decarbonization pathways are commensurate with global temperature goals of the Paris Agreement. A corporate-level ITR calculation is a counterfactual that projects hypothetical aggregate global warming if its emissions trajectory was representative of the global investment universe. ITR projects each company's cumulative emissions (Scopes 1, 2, and 3) by 2070, based on current emissions and emissions-reduction targets.[10] Each company's cumulative projected emissions are then compared to a GHG budget, specific to its industry and regional exposure. Companies with favorable ITR assessments include those with comprehensive near- and long-term net-zero targets and those with inherently low carbon footprints. **Figure 15** shows that our equity ITR (Scopes 1 and 2, Scopes 1, 2, and 3) and corporate-credit ITR (Scopes 1 and 2) are lower than those of their respective indexes, while our corporate-credit ITR (Scopes 1 and 2) is in line with its index. Many of the top contributors to ITR are oil, gas, and consumable-fuel companies, given their high Scope 3 emissions and low likelihood of incorporating Scope 3 emissions into a medium- or long-term target.

Figure 15

ITR of Wellington portfolios (degrees Celsius)



Data as of 31 December 2021. Source: Wellington Management. Please see the paragraph below for the assumptions and limitations associated with ITR.

Investors must consider the limitations and assumptions in MSCI's calculation of ITR. First, for companies that have not yet published decarbonization targets, cumulative emissions are assumed to grow 1% annually. We believe this is likely to be conservative, as pressure for companies to meaningfully reduce their emissions in absolute terms will continue to increase between now and 2070. Second, the inclusion of Scope 3 emissions is based on an estimated dataset from MSCI that ignores even robust corporate disclosure of Scope 3. While we expect the Scope 3 estimation model to be directionally appropriate at the industry level, it is unlikely to capture nuances between peer companies. Finally, ITR lacks transparency on company-specific GHG budget pathways for comparison against projected emissions. For this reason, is not possible to ensure that the ITRs of two companies from the same industry but in different regions are calculated by comparing their emissions trajectories to the same GHG budget.

PORTFOLIO-LEVEL CLIMATE SCENARIO ANALYSIS

We are able to utilize third-party scenario tools for total portfolio analysis; however, our investment teams are not currently managing portfolios to a particular scenario analysis metric. We believe data is lacking (specifically reliable Scope 3 emissions data for most issuers) for these models to be accurate. In addition, third-party providers lack sufficient transparency in their models on both climate and cost impacts for us to fully assess their accuracy. Nonetheless, we see value in analyzing the results we gain from these tools for other purposes.

USING SCENARIO-ANALYSIS METRICS

Climate value-at-risk, often abbreviated as Climate VaR or CVaR, is based on the principles of a standard value-at-risk calculation, which seeks to quantify the extent of possible financial losses over a specific time frame in present-value terms. Applied to climate change, CVaR attempts to make a direct connection between climate risks and the valuation of securities. Given that scenario analysis represents the risk of failing to execute against stated mitigation and resilience targets, CVaR measures a company's transition-risk exposure. We primarily use this metric, therefore, to sense-check our own research and inform engagement prioritization.

Using the MSCI CVaR portfolio-analysis tool, we select a range of transition-pathway scenarios and the average-physical-risk scenario. For the transition-risk component, we select the suite of scenarios from the integrated assessment model "REgional Model of Investment and Development" (REMIND). The Potsdam Institute for Climate Impact Research for the Network for Greening the Financial System (NGFS) developed REMIND to provide a common starting point for financial institutions to analyze climate risks. For the physical-risk component, the model provides the present value of the expected cost distribution consistent with RCP8.5, the representative concentration pathway published by the Intergovernmental Panel on Climate Change (IPCC), which best demonstrates the current trajectory or "business-as-usual" scenario.

We compare the outputs from the MSCI CVaR portfolio analysis tool with our internal research, which has a more

10 MSCI projects annual emissions through 2070 to align with the 2°C reference pathway. The Intergovernmental Panel on Climate Change (IPCC) Special Report on 1.5°C concludes greenhouse gas emissions need to be net zero by 2070 to limit warming to 1.5°C, consistent with net-zero carbon dioxide (CO2) emissions by 2050.

forward-looking orientation, to ensure holdings with a high CVaR have plans in place to effectively manage their climate-risk exposure. If companies lack a credible transition plan, we may adjust our engagement priorities.

APPLYING SCENARIO ANALYSIS FOR PORTFOLIO OVERSIGHT: CASE STUDY

The following case study of one of our global corporate-credit strategies shows how we use scenario analysis metrics as a tool for portfolio monitoring and oversight. We chose this strategy because, as a long-duration buy-and-hold approach, its CVaR is typically more pronounced than that of a short-duration strategy.

For diversified portfolios, we generally expect the Net Zero by 2050 (1.5°C Orderly) scenario to project the lowest overall CVaR. This is because climate policies are introduced immediately and gradually, which results in low physical risks. The Delayed Transition (2°C Disorderly) scenario projects, on average, higher value-at-risk from transition risks because policies are assumed to be enacted after 2030. As a result, those policies must be strong and call for significant changes. The Current Polices (3°C Hot House World) scenario assumes only currently implemented policies remain in place, leading to high physical risks despite lower transition risk.

The aggregate CVaR result is comprised of five primary components, including four transition-risk metrics and one physical-risk metric:

- Policy risk from direct emissions – Scope 1
- Policy risk from electricity use – Scope 2
- Policy risk from value chain – Scope 3
- Technology opportunities
- Physical risk from acute and chronic hazards

The first three components comprising Policy Risk VaR apply a carbon price to the emissions reduction required for each company to meet the selected transition-risk scenario. These cost projections do not consider a company's announced reduction targets or capital allocation.

As **Figure 16** shows, the high transition-risk CVaR results for the sample portfolio under the Delayed Transition scenario are related to the Policy CVaR (-1.6% for Scope 1 and -2.1% for Scope 2). This is predominantly because of the portfolio's utilities' holdings, which have high Scope 1 emissions. Digging deeper into the top contributor within utilities, our internal climate research and ESG research finds this company's decarbonization target and implementation strategy align with a below-2°C scenario. We therefore conclude that this Policy CVaR is unlikely to materialize. We expect to continue our ongoing engagement with this utility company to ensure it executes against its targets and encourage management to seek external validation of its targets from the SBTi.

The Technology Opportunities component projects future revenue, based on current and expected green revenues and patent value, is positive. While the portfolio-level result for Technology Opportunities is insignificant, certain underlying holdings show positive values, particularly capital goods and semiconductor companies. These results are intuitive, given the importance of intellectual property to the innovation of these industries and MSCI's patent-valuation methodology. Several utility companies also show positive values, thanks to the current share of renewable sources in their electricity-generation mixes. Because companies' stated targets are not incorporated into this assessment (and lesser importance of patents in the sector), we believe the future green revenue of utilities holdings is underestimated by this component of CVaR.

The Physical Risk component estimates the costs of business disruption and asset damage from acute and chronic events, based on location- and sector-specific vulnerability assessments. The results are consistently -0.2%, which we consider to be low/immaterial overall. The securities with the highest CVaR associated with extreme weather events are in the telecommunication services, transportation, and utilities industries. These results may be biased by the availability of a robust data set of locations of operations for these companies, relative to other industries.

We find consistent results when we apply our own analysis of physical risks to this sample portfolio, although our analysis highlights greater risk for certain oil and gas holdings. In one case, the Climate Research Team highlighted a company's operations in coastal regions, which were vulnerable to flooding. Our flood analysis relies on an elevation model proprietary to Woodwell. In another case, we highlighted a company's operations in the drought-prone Permian Basin. Here, our analysis relied upon a wider array of factors to arrive at drought and water scarcity (as opposed to MSCI's focus on river flow) to assess water scarcity.

Figure 16

Sample portfolio-level CVaR results

CVaR component	1.5°C (%AUM)	2°C (%AUM)	3°C (%AUM)
Policy - Scope 1	-0.1%	-1.6%	0.0%
Policy - Scope 2	-0.1%	-2.1%	0.0%
Policy - Scope 3	0.0%	-0.1%	0.0%
Technology opportunities	0.0%	0.0%	0.0%
Total transition risk	-0.2%	-3.8%	0.0%
Total physical risk	-0.2%	-0.2%	-0.2%
Total CVaR	**-0.3%**	**-4.0%**	**-0.2%**

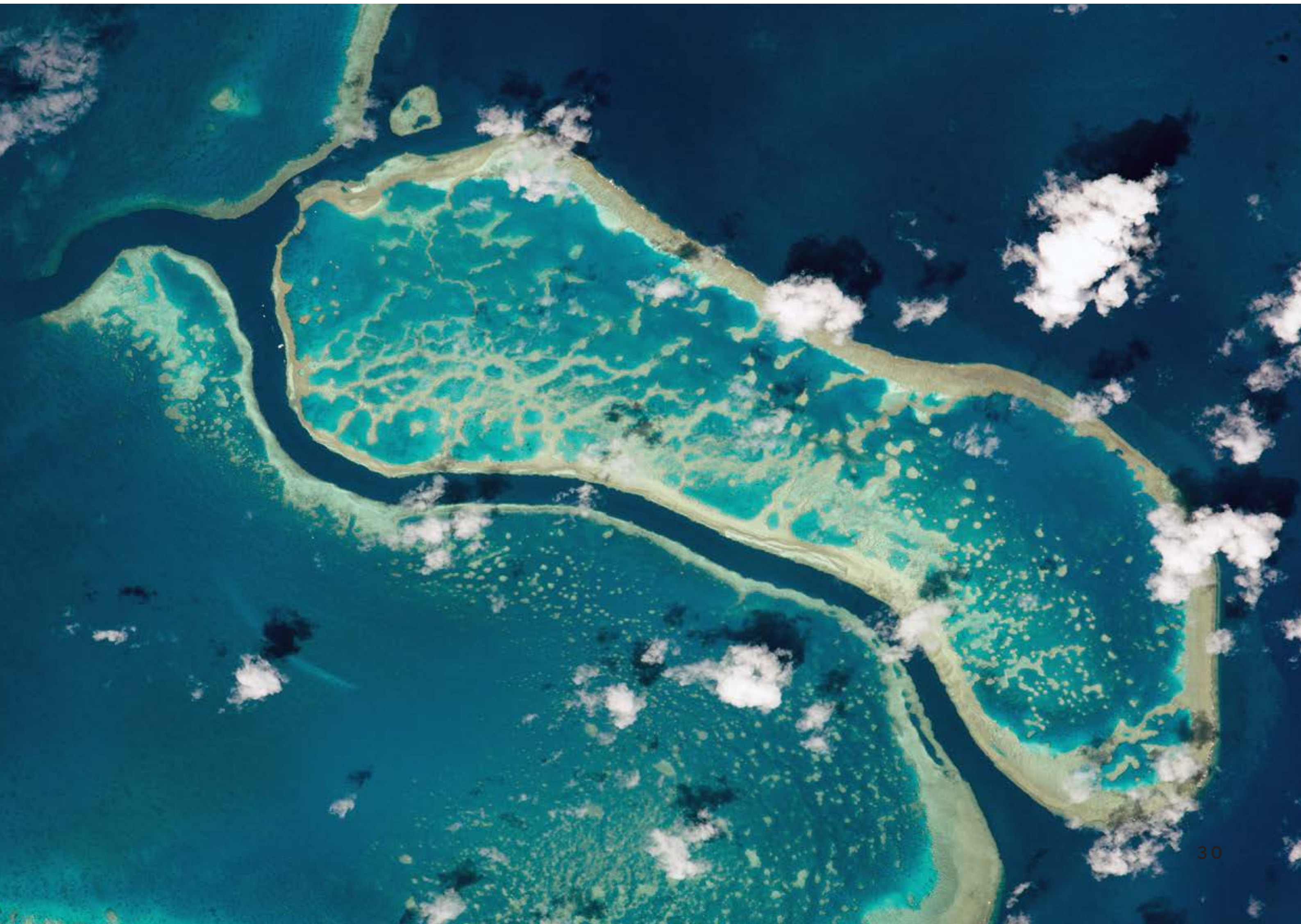
Data as of 31 December 2021. Sources: MSCI, Wellington Management.

We will continue to work with our climate research partners, Woodwell and the MIT Joint Program, and third-party providers to develop our own scenario-analysis capabilities to inform investment decisions and to enable us and our asset owner clients to meet regulatory expectations.

APPLYING SCENARIO ANALYSIS TO CAPITAL MARKET ASSUMPTIONS

In 2021, our Climate Research Team partnered with our Investment Strategy (iStrat) Team to develop 10-year capital market assumptions (CMAs). We evaluated and incorporated a range of climate transition assumptions and potential physical-risk impacts to determine which asset classes are most likely to be affected by climate change over the next decade. We elected to use the second iteration of the scenarios designed by the NGFS since these models do not include acute physical risk. We leaned heavily on our partnership with Woodwell to devise country-level assumptions for future extreme climate events (e.g., a 1-in-100-year precipitation event), the cost of associated damages and recoveries, and potential impact on real GDP. The implications of this climate-aware CMA process are more pronounced for equity than for fixed income. For equities, the adjustment to components of earnings-per-share growth forecasts included higher revenue growth because of the inflationary nature of transition policies and because of lower margins, as companies cannot pass the entire cost of the transition through to their customers.

Overall, the lower-margin assumptions, which offset higher nominal GDP growth, changed emerging market equities the most. For fixed income, we observe opposing forces affecting yields. On one hand, lower real-GDP estimates translate to lower real terminal yields. On the other hand, higher inflation estimates lead to higher nominal terminal yields, which lead to higher terminal yields, on average. For fixed income assets with a high duration, the boost in income is often insufficient to offset the additional capital losses from higher ending nominal yields. The overall impact of higher default probability, driven by the more challenging macro environment on our credit-loss matrix, is modest. We believe these assumptions will be important inputs into asset allocation decisions.



Section 5: Addressing our operational footprint

Reflecting our efforts to incorporate climate physical and transition risks into our investment decision making, we are also deeply committed to addressing the environmental impact of our corporate operations. Our internal measurement processes confirm our main sources of operational environmental impact are electricity use, business travel, and office waste. During 2021, we continued to take steps to reduce these impacts and mitigate the broader environmental footprint of our office buildings.

GROUNDBREAKING VIRTUAL POWER PURCHASE AGREEMENT

enel Green Power

In 2021, Enel Green Power North America and Wellington agreed to terms for a 10-year virtual power purchase agreement (VPPA) for an 11 MW portion of the energy delivered to the grid by Enel Green Power's Rockhaven wind project. The arrangement covers approximately 48 GWh of wind energy annually, and the associated renewable energy certificates (RECs) are equivalent to avoiding 30,000 tons of CO_2 from the atmosphere each year.

Under the agreed terms, we expect the clean energy contracted for Wellington to equal or exceed the electricity needs for all Wellington US corporate offices as well as residential electricity usage of our approximately 2,200 US employees. We believe the deal is an innovative model for corporate renewable energy buyers as it addresses the decentralized, post-COVID business operations with many workers shifting from office-based to remote work.

In addition to this PPA, the landlords of our Boston and London offices source 100% of our electricity from renewables. In London, we also installed more than 1,100 LED lights to help reduce our electricity usage in 2021.

MINDFUL BUSINESS TRAVEL

In 2021, we formed a "Future of Travel" working group, charged with adapting our approach to business travel in the context of sustainability and the learnings of the global pandemic. The working group launched internal communications to help employees more carefully consider the environmental impacts of their business travel and to encourage efficiencies. Because the circumstances of each business trip are unique, we rely on the professional judgment of each employee to assess the necessity and mode of travel. Particularly regarding air travel, we encourage employees to consider:

- Trip purpose and importance. Is there an opportunity to combine multiple trips? Can a local colleague represent you at a given meeting?
- Virtual meetings. Are there viable remote options? Are you achieving a balance between in-person and virtual collaboration?
- Mode. Is there a rail option? Have you considered the relationship between carbon emissions and class of service?

We also use technology to track our progress reducing emissions. Dashboards provide a transparent view of our travel data by department and route, and a new internal tool allows employees to easily compare emissions across travel options

Image source: Enel Green Power

and suggests ways to travel smarter. We believe business travel, approached mindfully, is crucial for developing our client and company relationships. We will continue to evolve our approach aiming for more efficient ways to serve our clients while seeking to reduce our carbon emissions.

REDUCING OFFICE WASTE

In support of the circular economy, we donate the firm's gently used technology equipment and recycle unusable devices. As we purchase new business technology, we prioritize the selection of environmentally friendly alternatives using less energy, with a smaller carbon footprint. As for software, we outsource the applications, infrastructure, and services from our own data centers to Amazon Web Services (AWS) data centers and co-location data centers connecting to AWS, reducing overall energy requirements and increasing performance.

We encourage our employees to make use of the recycling programs for paper, bottles, cans, batteries, and plastics in all of our buildings. For example, our London office's "single-stream" waste policy helps ensure all general waste gets recycled; in our Boston office, 94% of cutlery and packaging on offer in our company café is compostable.

We also use an on-demand print management system to reduce paper and toner waste through a limited number of high-capacity, multifunction devices. We reduce or eliminate hard-copy materials for internal and client/prospect meetings whenever possible.

EXPANDING RESPONSIBLY

As a growing company, we now lease approximately one million square feet of office space around the world. Our buildings' environmental ratings are key selection criteria. We seek spaces with Leadership in Energy and Environmental Design (LEED) certification or similar regional governmental designations, and we seek landlords who support and employ renewable energy and other positive environmental practices. As we build out our interior spaces, we also pursue energy-efficient designs, including open-plan seating, which use less space and energy per occupant. We use nontoxic paint and adhesives; energy-efficient heating, ventilation, and lighting systems; and water-saving plumbing fixtures and greywater collection systems.

ENVIRONMENTAL CERTIFICATIONS FOR WELLINGTON OFFICES

Boston:	US/LEED: 280 Congress Street = Platinum; 100 Federal Street = Gold
Chicago:	US/LEED = Gold
Hong Kong:	HK/BEAM7 = Platinum
London:	UK/BREEAM8 = Very Good
Milan:	LEED = Gold
San Francisco:	US/LEED = Gold
Singapore:	BCA Green Mark = Platinum
Sydney:	Australia/NABERS9 = 4.5 star
Toronto:	Canada/LEED = Gold

In 2021, we announced our commitment, beginning in 2023, to a long-term, 105,000 square-foot lease in Needham, Massachusetts, which will carry the LEED Zero Carbon Certification, making it the first office of its scale to achieve this status in the state. Collaborating with our landlord, Boston Properties, this buildout will include a deep energy retrofit, full electrification of gas-fired systems, heating, and ventilation. It also entails heating, ventilation, and air conditioning (HVAC) modernization, including advanced heat recovery and on-site renewable energy generation from a solar photovoltaic system designed to exceed our annual consumption. In addition, we have announced we will be ending our lease on our office at 100 Federal Street (currently LEED/US gold-rated), and will expand the space we lease in the LEED/US platinum-rated 280 Congress Street location.

MITIGATING THE IMPACTS OF EXTREME WEATHER EVENTS

We acknowledge the potential for business disruption caused by extreme weather events. Our offices in Boston, Hong Kong, Sydney, London, San Francisco, Singapore, and Tokyo are particularly susceptible to various climate-and environment-related risks, including power-supply disruptions and flooding. To mitigate the risk of disruptions, in recent years we have instigated various initiatives, including an AquaFence® in Boston which is an interlocking panel "fence" creating a portable four-foot (1.2-meter) buffer around the building's perimeter. In Tokyo, Japan, a city susceptible to earthquakes, we relocated our office to the lower levels of a new property featuring earthquake-resistant structural design.

During 2021, we leveraged our CERA tool to assess the climate resilience of our office ecosystems and key service providers. We shared the results and conclusions of this analysis with leaders of a number of our operations and infrastructure teams, including those responsible for workplace services, business continuity, procurement, and third-party risk.

DISCLOSING OUR PHYSICAL LOCATION DATA

In line with our call on companies to disclose their physical location data, we intend to disclose our owned, leased, or otherwise operated physical assets in a publicly accessible format (**Figure 17**).

Figure 17

Wellington's physical location data

Street address	Country	Facility	Ownership	Coordinates	Materiality
280 Congress Street, Boston, Massachusetts, 02110	United States	Office building	Leased	42.353516767286365 latitude -71.05253903273685 longitude	46% of global employees
100 Federal Street Boston, MA 02110	United States	Office building	Leased	42.35523508990214 latitude -71.05603427479248 longitude	20% of global employees
222 West Adams Street, Suite 2100, Chicago, Illinois, 60606	United States	Office building	Leased	41.87967291850019 latitude -87.63469154411777 longitude	1% of global employees
100 Campus Drive, Marlborough, Massachusetts, 01752	United States	Office building	Leased	42.325285123258574 latitude -71.5850687171218 longitude	7% of global employees
4 Radnor Corporate Center, Suite 500, Radnor, Pennsylvania, 19087	United States	Office building	Leased	40.049001411126355 latitude -75.35770764415707 longitude	1% of global employees
4 Embarcadero Center, Suite 2610, San Francisco, California, 94111	United States	Office building	Leased	37.79550469696568 latitude -122.39625178838118 longitude	1% of global employees
Exchange Tower, 130 King Street West, 18th Floor, Toronto, ON M5X 1E3	Canada	Office building	Leased	43.64875311300009 latitude -79.3833466305852 longitude	<1% of global employees
17F, Two International Finance Center, 8 Finance Street	Hong Kong Special Administrative Region of the People's Republic of China	Office building	Leased	22.28524817233272 latitude 114.15920991323107 longitude	4% of global employees
Unit 822, Level 8, International Finance Center Tower, 2 8 Century Avenue, Pudong District, Shanghai, 200120	China	Office building	Leased	31.23771299772283 latitude 121.50214974602783 longitude	<1% of global employees
8 Marina Boulevard, #03-01, Tower 1, Marina Bay Financial Center, 018981	Singapore	Office building	Leased	1.2801268484023878 latitude 103.85395702479049 longitude	3% of global employees
Palace Building 7F, 1-1-1 Marunouchi, Chiyoda-ku, Tokyo 100-0005	Japan	Office building	Leased	35.68532510967304 latitude 139.7615711827426 longitude	3% of global employees
Bockenheimer Landstraße 43-47, 60325, Frankfurt am Main	Germany	Office building	Leased	50.11729920644733 latitude 8.665131754230236 longitude	1% of global employees
80 Victoria Street, London, SW1E 5JL	United Kingdom	Office building	Leased	51.49719414805737 latitude -0.13958625922653942 longitude	12% of global employees
33 avenue de la Liberté, L-1931	Luxembourg	Office building	Leased	49.60439819142996 latitude 6.130880971409155 longitude	1% of global employees
Limmatquai 92, 8001 Zurich	Switzerland	Office building	Leased	47.374203851981754 latitude 8.543196913681628 longitude	<1% of global employees
Level 17, 126 Phillip Street, Sydney, NSW 2000	Australia	Office building	Leased	-33.86700645122414 latitude 151.21161834245913 longitude	1% of global employees
Via Dante 7, 20123, Milan	Italy	Office building	Leased	45.466626811449046 latitude 9.184266596443248 longitude	<1% of global employees
Exposure to colocations					
1 Summer St, Boston, MA, 02110	United States	Data center	Leased	42.354857748661864 latitude -71.06027080177864 longitude	No employees
15 Pioneer Walk, 627753	Singapore	Data center	Leased	1.3221111005077115 latitude 103.69525299780426 longitude	No employees
555 Scherers Ct, Columbus, OH, 43085	United States	Data center	Leased	40.11588015152274 latitude -83.00236901532 longitude	No employees
12100 Sunrise Valley Dr, Reston, VA, 20191	United States	Data center	Leased	38.95077936576078 latitude -77.36446744233018 longitude	No employees
800 Secaucus Rd, Secaucus, NJ, 07094	United States	Data center	Leased	40.77863999516403 latitude -74.07217292879902 longitude	No employees
1905 Lunt Ave, Elk Grove Village, IL, 60007	United States	Data center	Leased	42.00127298368297 latitude -87.95492777480031 longitude	No employees
SL1 4AX	United Kingdom	Data center	Leased	51.52230393676966 latitude -0.6291364981284363 longitude	No employees
21715 Filigree Ct, Ashburn, VA, 20147	United States	Data center	Leased	39.016677369721116 latitude -77.45932607486368 longitude	No employees

BECOMING CARBON NEUTRAL

In our 2020 Sustainability Report, we set out a comprehensive carbon-footprint measurement covering our global operational emissions from 2014, providing a baseline. We have since used this data (encompassing Scopes 1, 2, and 3 emissions) to develop strategies to reduce our operational carbon footprint, in addition to investing in credible offsets for remaining emissions to meet our target to be operationally carbon neutral by the end of 2022 (**Figure 18**).

We take a multifaceted approach to ensure we purchase high-quality offsets aligned with our broader sustainability goals. As available tools and technologies improve and become economical at scale, we expect to evolve our approach. In 2021, we purchased enough offsets to become operationally carbon neutral relative to our 2020 emissions. We also chose to align the volume of offsets with our 2019 — pre-pandemic — travel emissions, to reflect our intention to go beyond carbon neutrality. This included purchasing offsets from CarbonCure™ whose innovative methods capture carbon dioxide in concrete with high measurability, additionality, and permanence.

We will also actively support the development of the carbon offset market, incorporating the Oxford Principles for Net Zero Aligned Carbon Offsetting, including selecting high-quality offsets, modeling best practices, supporting transparency, and aligning with our clients and industry peers. Learn more in "Toward carbon neutrality: our approach to carbon offsets," published on our website.

Figure 18

Carbon offset projects

Project name	Project type	Country	Methodology or protocol	Third-party standard
Maisa REDD+ Project	Forestry	Brazil	VM0015	Verified Carbon Standard (VCS)
Hestian	Cookstoves	Rwanda & Malawi	GS TPDDTEC v 1	Gold Standard
CarbonCure's sustainable concrete solution	CO2 utilization in difficult-to-abate sectors	US/ Canada	VM0043	VCS

OUR OPERATIONAL CARBON FOOTPRINT

Our 2021 company emissions (**Figure 19**) have been calculated in line with the World Resources Institute and World Business Council for Sustainable Development's Greenhouse Gas Protocol Corporate Accounting and Reporting Standard (revised). The emissions reported are for Wellington's financial year and relate to operations in our global offices.

The boundaries of the GHG inventory are defined using the operational control approach and cover the emissions for which the company is responsible. Emissions for previous years are retrospectively adjusted as more accurate data becomes available.

Figure 19

Wellington's carbon emissions

Metric tons CO2e	2017	2018	2019	2020	2021	% change 2021 vs 2014 (baseline)	% change 2021 vs 2020 (year over year)
Scope 1[a]	1,282	1,276	1,304	1,161	1,134	-12%	-2%
Scope 2[b] market-based	3,639	3,823	3,307	2,292	1,782	-56%	-22%
Scope 2[b] location-based	6,673	6,887	6,603	5,163	3,991	-46%	-23%
Total market-based own source (Scopes 1 and 2)	4,921	5,099	4,611	3,453	2,916	-46%	-16%
Scope 3[c]	9,448	10,728	11,445	1,393	1,057	-90%	-24%
Total market-based emissions (Scopes 1, 2, & 3)	14,370	15,827	16,055	4,848	3,973	-75%	-18%
Headcount*	2,926	3,059	3,142	3,108	3,188	13%	3%
MTCO2e/person own source	1.7	1.7	1.5	1.1	0.9	-52%	-18%
MTCO2e/person Scope 3	3.2	3.5	3.6	0.4	0.3	-92%	-36%
MTCO2e/person**	4.9	5.2	5.1	1.6	1.2	-78%	-20%
Company footprint (square meters)	90,019	90,088	91,101	86,317	83,954	-7%	-3%
MTCO2e/sq m own source	0.055	0.057	0.051	0.040	0.035	-42%	-13%
MTCO2e/sq m Scope 3	0.105	0.119	0.126	0.016	0.013	-89%	-22%
MTCO2e/sq m**	0.160	0.176	0.176	0.056	0.047	-73%	-16%

a Scope 1 emissions sources include the on-site combustion of natural gas for space heating, the combustion of diesel for backup generators, and fugitive refrigerants from HVAC systems.

b Scope 2 emissions sources include the purchase of electricity at all locations and the purchase of steam at one site for space heating. This table calculates emissions using a market-based approach and a location-based approach. Purchased electricity emissions factors were adjusted in 2021 to account for current carbon intensity of the power grids in various locations where Wellington operates. European emissions factors in 2021 were updated to the Association of Issuing Bodies data source and calculated under national level average grid mix factors (location-based) and national level residual grid mix factors (market-based). Emissions were down slightly year over year, with electricity purchases declining and the carbon intensity of power grids falling vs previous reporting cycles.

c Scope 3 emissions sources primarily include business travel (air and rail) for all reported years. Starting in 2019 when Wellington completed its cloud transition, the company elected to include emissions related to cloud services and colocation data centers. Most of the decline in Scope 3 emissions resulted from the impact of the global pandemic on Wellington's travel-associated emissions.

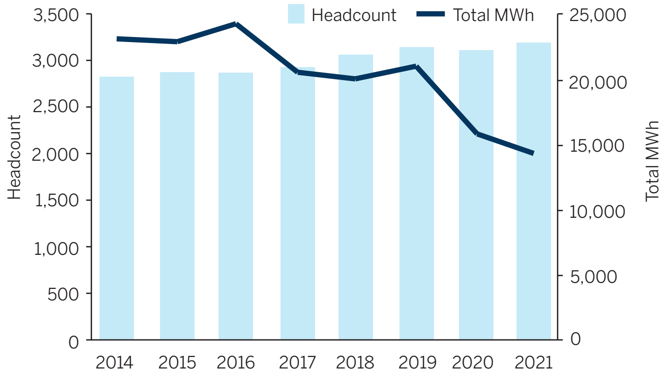
* Headcount figures include all full-time and contingent workers with an assigned seat location in one of our leased facilities. We track headcount monthly and take an average across 12 months to reflect office occupancy more accurately.

** Key performance indicators measuring emissions totals under market-based accounting methods.

We reduced our total Scopes 1, 2, and 3 operational emissions by 18% between 2020 and 2021 and, since setting our benchmark in 2014, these emissions have fallen by 75%. We are mindful that there is likely to be an increase in our emissions in 2022 relative to 2020/2021, given some renewed business travel. We continue to actively consider ways of minimizing our emissions in 2022 and beyond. **Figure 20** shows our progress in reducing our electricity usage. We have managed to bring usage down, despite increasing headcount.

Figure 20

Wellington's electricity usage



Next steps and closing statement

We believe it is our fiduciary duty to understand how climate risks and opportunities may impact client portfolios. We will strive to maintain our position as a leader in the industry on climate research and lean into our seat of influence as shareholders to encourage our portfolio companies to adopt best practices in climate-risk management. Specific areas of focus follow.

DEEPENING OUR RESEARCH

Our 2022 climate research agenda encompasses physical and transition risks, leveraging our partnerships with Woodwell and the MIT Joint Program, respectively. With Woodwell, we plan to dive deeper into biodiversity and the effects of physical risks on supply chains. We will pursue several new country case studies, in which we seek to better understand each country's physical climate risks and opportunities. We will continue to enhance CERA by integrating both a higher-resolution data set and physical climate projections for multiple Representative Concentration Pathway (RCP) scenarios. With the MIT Joint Program, we expect to focus on knowledge building, expanding our understanding of transition risk in high-emitting sectors, as well as climate policies and their potential impact on companies across sectors.

We expect to study how internally developed climate CVaR calculations may be applied to certain sectors, and we will evaluate key climate scenario-analysis assumptions. We expect that the resulting insights from this research will deepen our engagements.

INTEGRATING CLIMATE RISKS AND OPPORTUNITIES INTO PORTFOLIOS

The Climate Research Team will continue to provide ongoing support across the investment platform, providing education sessions and highlighting potential capital market implications of scientific and policy research with investors around the firm. It will continue to partner with investment teams and industry analysts to conduct company- and portfolio-level evaluations on transition risk, net-zero alignment, and physical climate risk. We expect these collaborative assessments will uncover additional compelling investment opportunities in companies providing products and services to address mitigation and adaptation needs. In addition, the Climate Research Team will lead ongoing enhancements of our climate tools and data sets.

ADVANCING ADOPTION OF DECARBONIZATION PATHWAYS

Through our participation in numerous net-zero industry initiatives, we will continue to collaborate with other signatories to define benchmarking criteria, develop best practices for decarbonization methodologies, and shape industry standards. In addition, in 2022 we expect to increasingly submit our input, informed by our climate science and industry expertise, to SBTi as the organization solicits feedback via public consultations prior to publishing new guidance for sector-specific target-setting criteria. We believe the direction of SBTi guidance is worth monitoring, given our focus on bottom-up measurement of targets. Finally, we will deepen our relationships with existing third-party data providers, providing feedback and suggesting enhancements for datasets, and identify new providers for a wider range of asset classes.

MAKING PROGRESS TOWARD NET ZERO

We will continue to engage with and educate our clients on the financial materiality of climate change. We aim to help our clients assess the potential portfolio consequences of net-zero objectives and formalize portfolio decarbonization expectations to that end, where appropriate. In all cases, we will pursue net zero in nonconcessionary ways, aligned with client objectives and investment teams' philosophy and processes. We also will be improving portfolio reporting to clients as a means of driving greater transparency and accountability. As for portfolio companies, we will engage to encourage improvement on disclosure and setting science-based emissions-reduction targets. Finally, we will report periodic updates on net-zero-aligned AUM annually and announce AUM milestones when reached.

CLOSING STATEMENT

Wellington's founding partners instilled the notion of sustainability and stewardship of clients' assets, which continues to be a guiding principle for our firm. Our approach to climate integration is highly consistent with these views — we aim to uncover mispriced risks and opportunities tied to climate change and incorporate these organically into our philosophies and processes through data, research, engagement, and active management. We appreciate the opportunity to share our commitment and evolution in this report.

For additional information, please visit us online:

wellington.com

wellingtonfunds.com

linkedin

Climate investing is qualitative and subjective by nature, and there is no guarantee that the criteria utilized, or judgment exercised, by Wellington will reflect the beliefs or values of any one particular investor. Climate investing norms may differ by region. There is no assurance that any climate investing strategy and techniques employed will be successful.

Please refer to the Important disclosures at the end of this report.

Appendix 1. TCFD disclosure index

Below are cross-references to where the recommended disclosures can be found throughout this report.

Appendix 2. Sample strategies' carbon footprints

The strategies below are a subset of our strategies offered by Wellington Management and represent the largest long-only strategies (by AUM), with carbon data coverage greater than 50%, classified based on their investable universe. Portfolio-level emissions analytics are available to all portfolio managers but may or may not be used in security selection and portfolio construction.

	Carbon Data Coverage[1] %	Eligibility[2] %	WACI Scope 1+2[3] T CO2e/ $M Sales	WACI Scope 1+2+3[3] T CO2e/$M Sales	Carbon Footprint Scope 1+2[4] T CO2e/$M Invested	Carbon Footprint Scope 1+2+3[4] T CO2e/$M Invested	ITR Scope 1+2+3[5] °C	SBTs Target[6] Set %	SBTs targets set + committed[6] %
EQUITY									
Global									
Global Quality Growth	97.6	99.6	21.6	353.1	3.8	92.8	2.0	30.1	39.2
Global Impact	98.4	94.8	47.0	464.6	12.7	134.8	2.1	23.1	28.6
Global Research Equity	99.1	97.8	53.9	504.3	17.6	202.7	2.2	29.0	41.9
Index: MSCI All Country	99.8	100.0	151.5	804.0	53.0	317.6	2.8		
US									
US Research Equity	99.0	99.1	57.3	367.7	17.2	96.5	1.9	20.8	34.5
US Quality Growth	98.8	99.4	16.6	259.1	4.0	79.0	1.8	27.6	39.6
Index: S&P 500	99.9	100.0	131.1	667.9	32.8	203.0	2.6		
Emerging markets									
Emerging Markets Research Equity	98.4	97.3	212.3	1370.9	100.9	589.8	3.9	4.9	9.2
Emerging Markets Local Equity	95.5	95.9	163.0	635.3	38.8	155.6	2.7	4.0	7.5
Index: MSCI Emerging Market	99.4	100.0	328.9	1342.5	141.1	625.1	3.7		
Asia Technology	97.8	99.1	78.8	500.3	17.7	137.6	2.4	12.8	16.2
Index: MSCI AC Asia Pacific Information Technology	99.9	100.0	109.5	523.5	27.0	195.4	2.2		
Europe									
Strategic European Equity	99.5	99.2	24.1	342.6	10.9	148.7	1.9	28.4	48.4
Focused European Equity	100.0	98.5	152.5	679.2	99.1	395.5	2.3	47.9	59.6
Index: MSCI Europe	99.8	100.0	122.6	830.1	68.4	452.4	2.7		
Pan European Small Cap Equity	89.5	98.5	41.7	545.6	14.1	165.8	2.5	16.0	32.8
Index: MSCI Europe Small Cap	97.2	100.0	106.9	655.0	65.6	388.5	2.5		
Sector									
Global Health Care	96.4	98.2	20.5	327.5	3.7	63.2	1.7	18.0	24.5
Index: MSCI World Healthcare	100.0	100.0	20.2	331.1	3.9	73.1	1.6		
FinTech	97.1	99.7	6.7	142.3	0.6	11.8	1.5	17.8	21.4
Enduring Assets	100.0	98.1	698.7	1669.0	170.8	511.8	2.9	29.2	36.3
Index: MSCI All Country	99.8	100.0	151.5	804.0	53.0	317.6	2.8		

	Carbon Data Coverage[1] %	Eligibility[2] %	WACI Scope 1+2[3] T CO2e/ $M Sales	WACI Scope 1+2+3[3] T CO2e/$M Sales	Carbon Footprint Scope 1+2[4] T CO2e/$M Invested	Carbon Footprint Scope 1+2+3[4] T CO2e/$M Invested	ITR Scope 1+2+3[5] °C	SBTs Target[6] Set %	SBTs targets set + committed[6] %
FIXED									
Credit									
Global Credit Plus	94.7	80.9	129.6	595.4	40.6	234.9	2.5	22.5	37.1
Index: BBG GAgg Cr-fin US/EU/UK 1%	97.0	99.0	212.3	967.9	64.4	419.5	2.9		
Global Credit Buy and Maintain	92.0	84.8	222.9	913.7	58.9	350.4	2.5	23.6	35.7
High yield									
Global High Yield Bond	67.3	97.3	226.3	1499.0	97.7	804.5	4.0	7.5	11.1
Index: ICE BofA Gbl HY Constrain	80.4	99.3	371.8	1852.5	171.9	942.2	3.6		
Higher Quality Global High Yield Bond	71.9	96.4	244.0	1613.9	99.8	799.4	4.0	7.9	11.5
Index: ICE BofA Glob HY BB-B Constr	83.3	99.3	350.1	1847.2	172.0	920.2	3.6		
Euro High Yield Bond	55.7	97.1	126.4	909.0	30.9	426.8	2.9	10.0	19.4
Index: BBG US Aggregate	98.7	99.5	311.1	1105.0	57.7	389.3	3.0		
Credit									
Multi-Sector Credit	55.1	58.6	326.8	2014.4	105.1	729.4	3.6	4.3	7.3
Responsible Values Multi-Sector Credit	52.3	55.2	145.7	612.8	53.5	278.4	2.7	4.7	9.1

All calculations are scaled such that covered securities are reweighted to 100% market value; this means that covered holdings have an outsized impact on the portfolio. Portfolios with lower coverage can be especially influenced by outliers. Data may not be representative of the actual portfolio in cases where coverage of carbon data availability or eligible securities is between 50% and 75%.

All data is as of 31 December 2021. SBTs are validated by SBTi. All other data is sourced from MSCI.

1 Carbon data coverage is represented as a % of carbon eligible securities, which may be less than the total market value of the portfolio.

2 Eligibility is based on the state of carbon data availability and includes only corporate holdings.

3 WACI: Carbon footprint reporting accounts for Scope 1 and 2 GHG emissions, as well as Scope 1, 2, and 3 emissions, and is expressed in carbon dioxide equivalents. The WACI figure for each strategy is calculated by rescaling exposures based on available emissions data and therefore may not be fully representative of the strategy's emissions.

4 Carbon Footprint: Emissions financed per US$1 million invested in the mandate. This metric is calculated by summing the result of '% Enterprise value incl cash financed X Emissions' for each holding, and then dividing by the portfolio's total market value.

5 Implied Temperature Rise (ITR): ITR is a forward-looking metric, expressed in degrees Celsius, and is designed to show the alignment of companies with global temperature goals, between 1.5°C and 2°C. It aims to quantify the alignment of a company's activities against pathways commensurate with future temperature goals, and measures the global warming if the rest of the investment universe acts the same way as a single company. Each company's cumulative emissions by 2070 (Scopes 1, 2, and 3) is projected based on their current emissions and emissions reduction targets, where the current Scope 3 emissions reflect MSCI's fully estimated Scope 3 data. Each company's cumulative projected emissions are then compared to a GHG budget specific to its industry and regional exposure when calculating its ITR. ITR data availability is determined by the % of eligible carbon MV with ITR and EVIC data.

6 Science Based Targets (SBTs): SBTs are 10-year targets aligned with limiting global warming to 1.5°C, include Scope 1, 2, and material Scope 3 emissions, and must be achieved through direct action in operations/value chain. Three consecutive SBTs would meet net zero by 2050. Eligibility is currently based on exposure to long-only, direct corporate holdings and excludes look-through to pools. SBTs' data availability is determined by the exposure of the % of eligible MV. Targets Set indicates the company's target has been validated to be comprehensive, robust, and aligned with approved methodologies, while Committed indicates the company is in the process of setting its target and seeking review from SBTi.

NOTES ON METHODOLOGY AND CURRENT LIMITATIONS OF CARBON FOOTPRINTING

Since emissions are self-disclosed and audited inconsistently, there is a lack of uniformity from one company to another. Estimation methodologies by third parties are imperfect because they are based on assumptions about business operations, peers, and industry averages. There is also a time lag associated with disclosures as companies disclose emissions at the end of each fiscal year, so the holdings date does not exactly match the emissions year.

Where a strategy incorporates transition-risk management into its P&P, we currently focus portfolio construction and target-setting on Scope 1 and 2 emissions, where we have the most confidence in the data due to corporate disclosure. However, consideration of Scope 3 is important because it often provides the strongest tie to business strategy. For example, Scope 3 downstream (use of sold products) provides the strongest tie to top-line opportunities, which allow us to differentiate companies' current exposure to demand for low-carbon products, such as the electric vehicle (EV) offering of automobile manufacturers.

Because carbon emissions are a moment-in-time snapshot, they fail to capture companies' future transition strategy. Companies with forward-looking reduction targets may nonetheless seem carbon-intensive today. We encourage portfolio managers to consider multiple metrics, including forward-looking ESG ratings, capital spending plans, and science-based target commitments, when analyzing portfolios.

At cyclical companies, carbon intensity can fluctuate dramatically without a change in business processes, as revenue is the denominator in the formula. Currency translation for companies reporting in currencies other than the US dollar can also impact revenue-based intensity figures. When the ESG Research Team looks at individual sectors, they tend to use units of production rather than revenue, especially for cyclical industries.

Additional challenges arise when footprinting corporate credit strategies. Propagation from a parent issuer to a subsidiary issuer is done to improve coverage of the portfolio. However, if a parent issuer is the disclosing entity, this parent may or may not be representative of the subsidiary issuer owned. This may be resolved over time as investors come to expect enhanced emissions disclosures directly from the issuing entity.

Please refer to the discussion and footnotes within Section 4, Metrics and targets, for further consideration of our full range of climate-related metrics.

Appendix 3. 2021 list of company engagements

During 2021, we held 17,500 meetings, thousands of which were hosted by our equity, credit, and ESG analysts, as well as portfolio managers. As the vast majority of our company meetings each year discuss long-term strategic business issues, we recently introduced a complementary engagement-tracking tool for all our investment professionals. This tool provides an interface to allow users to add topics, feedback, and outcomes of recently attended company meetings whereby investors can enter engagement information associated with each meeting they attend. In particular, they are tracking the feedback they provided to the company and the outcome of the engagement, in terms of how it influenced an investment decision and/or how the company is taking action based on the feedback.

In 2021, we engaged with over 600 companies in 48 countries where climate change was among the discussion topics. With the aim of ongoing communications, we often meet with companies multiple times throughout the year.

COMMUNICATION SERVICES

Altice Europe
CDG
Cellnex Telecom
China Telecom
Comcast
Liberty Global
Moody's
Paramount Global
SEEK
Switch
Telecom Italia
T-Mobile US
Trip.com
Wolters Kluwer
ZipRecruiter

CONSUMER DISCRETIONARY

Adient
Amazon.com
Beazer Homes USA
Caesars
Carnival
Carnival Industrial
Coats
Compagnie des Alpes
Compass
Deckers Outdoor
Detsky Mir
Etsy
Ford Motor
General Motors
H & M Hennes & Mauritz
Home Depot
Honda Motor
Host Hotels & Resorts
Huazhu
Industria de Diseno Textil
Kering
Kia
La-Z-Boy
Lear
Lennar
Lowe's
Meituan
Midea
PAL
Pool
PulteGroup
RealReal
SkiStar
Starbucks
Sundrug
Terminix
TJX
VF
Vivo Energy
Volkswagen
WESCO International

CONSUMER STAPLES

Anheuser-Busch
Budweiser Brewing
Bunge
Carrefour
China Feihe
Coca-Cola
Coca-Cola Europacific Partners API Pty
Coca-Cola Europacific Partners
Colgate-Palmolive
Constellation Brands
CP ALL
Darling Ingredients
Diageo
Estee Lauder
Freshpet
General Mills
Jiangsu Hengshun Vinegar Industry
Kellogg
Kroger
Kweichow Moutai
Lamb Weston
Mondelez International
Monster Beverage
Nestle
Nomad Foods
Philip Morris International
Procter & Gamble
Sanderson Farms
Swedish Match
Sysco
Tingyi Cayman Islands
Unilever
United Breweries
Viscofan
Walmart
Yihai International

ENERGY

Aker
Anadarko Petroleum
ARC Resources
Baker Hughes
BP
Brigham Minerals
California Resources
Cheniere Energy
Chevron
China Gas
Cimarex Energy
CNOOC
Concho Resources
ConocoPhillips
Coterra Energy
Crestwood Equity Partners
Devon Energy
Diamondback Energy
DMC Global
DT Midstream
Ecopetrol
Enbridge
Energean
Energy Transfer
Energy Transfer Operating
Eni
EnLink Midstream
EOG Resources
Equinor
Expro Group
Extraction Oil & Gas
Exxon Mobil
Frontera Energy
Galaxy Pipeline Assets Bidco
Galp Energia SGPS
Gazprom PJSC
Gulfport Energy Operating
Halliburton
Hess
Inpex
Japan Petroleum Exploration
Karoon Energy
Kinder Morgan
Kosmos Energy
LUKOIL
Lundin Energy
Marathon Oil
Marathon Petroleum
Murphy Oil
Neste
Noble Energy
NOV
Novatek
Occidental Petroleum
ONEOK
Ovintiv Exploration
Pembina Pipeline
Petroleo Brasileiro
Phillips 66
Pioneer Natural Resources
ProPetro
QEP Resources
Raizen
Renewable Energy
Repsol
Rosneft Oil
RPC
Schlumberger
Shell
Southwestern Energy
Storm Resources
Subsea 7
Suncor Energy
Targa Resources
TC Energy
Tenaris
TotalEnergies
Valaris
Valero Energy
Whiting Petroleum
Williams Companies
Wintershall Dea
Worley
YPF

FINANCIALS

Aegon
Alleghany
Allianz
Allstate
American Financial Group

American International Group
ArcLight Clean Transition
Assured Guaranty
Aviva
Avolon Holdings Funding
AXA
Banco Bilbao Vizcaya Argentaria
Banco Santander
Bank of Ireland
Bank of Nova Scotia
Bankia
Barclays
BAWAG Group
Beazley
Belfius Bank
Blackstone
BNP Paribas
Brookfield Asset Management
CaixaBank
Castellana Properties
China Construction Bank (Asia)
China Vanke
Chubb
Cincinnati Financial
Citigroup
Close Brothers
CNO Financial
Commerzbank
Credit Suisse
DBS
Deutsche Bank
Erste Group Bank
Everarc H
Everest Re
FirstRand
FleetCor Technologies
Genworth Mortgage Insurance Australia
Hamburg Commercial Bank
Hannover Rueck
Hartford Financial Services
HSBC
Huron Consulting
Ichigo
IFS Capital
ING
Insurance Australia Group
Intact Financial
JPMorgan Chase
Just
KBC
Legal & General
Lincoln National
Lloyds Banking
London Stock Exchange
M&G
Mediobanca Banca di Credito Finanziario
Metro Bank
MGIC Investment
Mitsubishi UFJ Financial Group
Molina Healthcare
Nasdaq
Nationwide Building Society
NatWest
Paradigm Homes Charitable Housing Association
Permanent TSB Group
Phoenix Group
Playa Hotels & Resorts
Principality Building Society
PROG
Progressive
Raiffeisen Bank International
Reinsurance Group of America
RenaissanceRe
Santander UK
Skipton Building Society
Societe Generale
Standard Chartered
Sunlight Financial
Swiss Re
Texas Capital Bancshares
Travelers
UBS
UnitedHealth
Voya Financial
W R Berkley
White Mountains Insurance
Xenia Hotels & Resorts
Zurich Insurance

HEALTH CARE

Almirall
AstraZeneca
Avanos Medical
Baxter International
Boston Scientific
Danaher
Eisai
Encompass Health
GlaxoSmithKline
HCA Healthcare
Hologic
Illumina
Incyte
Nakanishi
ObsEva
Otsuka
Pfizer
Sanofi
Seagen
Smith & Nephew
Teva Pharmaceutical Industries
Zai Lab

INDUSTRIALS

AAON
Advanced Drainage Systems
Aena
AerCap
Alaska Air
Allied Universal
Allison Transmission
Alstom
AP Moller - Maersk
Argan
Arrival
Atlas Copco
Befesa
Beijing Oriental Yuhong Waterproof Technology
BeNext-Yumeshin
Bloom Energy
Boskalis Westminster
BrightView
Builders FirstSource
Bureau Veritas
BYD
Canadian National Railway
Caterpillar
Cemex
China Glass
China Resources Cement
Cie de Saint-Gobain
Cie Generale des Etablissements Michelin
Continental Building Products
Copart
Core & Main
Coventry Group
Daikin Industries
Deere
Delta Air Lines
Deutsche Lufthansa
DMCI
Dover
DP World/United Arab Emirates
East Japan Railway
Ecopro
Electrocomponents
Energy Recovery
EnerSys
Escorts
Evoqua Water Technologies
FANUC
First Student Bidco
Fluidra
Franklin Electric
GATX
Gaztransport Et Technigaz
Generac
General Dynamics
Genuit
Giken
Graphic Packaging
Great Lakes Dredge & Dock
Havells India
Hazama Ando
Hexcel
Honeywell International
Hubbell
IDEX
IHI
Illinois Tool Works
Indika Energy
Insteel Industries
International Container Terminal Services
Japan Airlines
JB Hunt Transport Services
JinkoSolar
Jonjee Hi-Tech Industrial and Commercial
Kingspan Group
Kuehne + Nagel International
Kyudenko
Lennox International
Li Auto
Lockheed Martin
Loomis
Martin Marietta Materials
Max Co
Mercedes-Benz
Mitsubishi
National Central Cooling
National Instruments
NexTier Oilfield Solutions
Nikola
NIO
Nissin Electric
Niu Technologies
Norfolk Southern
Northrop Grumman
Northwest Pipe
nVent Electric
Obara
Orion
PACCAR
Pentair
PGT Innovations
Radiant Logistics
Raytheon Technologies
Recruit
Rentokil Initial
Republic Services
Rockwool International
Rollins
Ryanair
SAN Miguel Industrias Pet
Schneider Electric
Shin-Etsu Polymer
Signify
SM Investments
Smith & Wesson Brands
SolarEdge Technologies
Spirit Airlines
SPX Flow
Stantec
Stem
Sulzer
Sunrun
TaiSol Electronics
Tervita
Tetra Tech
Tidewater Midstream and Infrastructure
Trane Technologies
Uber Technologies
Union Pacific
United Airlines
United Parcel Service
Vaisala
Verallia
Vestas Wind Systems
Vinci
Vulcan Materials
Washtec
Watsco
Weichai Power
WSP Global
Wuxi Lead Intelligent Equipment
Xylem

INFORMATION TECHNOLOGY

Accenture
Analog Devices
Apple Inc
Applied Materials
Argo Graphics
ASML
Aspen Technology
Autodesk
Badger Meter
BE Semiconductor Industries
Bentley Systems
Block
Canadian Solar
Capgemini
Chindata Group
Elecom
ESPEC
Fidelity National Information Services
First Solar
Infineon Technologies
Intel
Intuit
Itron
KLA
Koh Young Technology
Kyocera
Lam Research
Microsoft
Mimasu Semiconductor Industry
MKS Instruments
Nexi
NEXTDC
ON Semiconductor
Oracle
Seiko Epson
ServiceNow
Skyworks Solutions
SUMCO
SunPower
Taiwan Semiconductor Manufacturing
Texas Instruments
Tokyo Electron
Tong Hsing Electronic Industries
WEX

MATERIALS

Agnico Eagle Mines
Air Products and Chemicals
Akzo Nobel
Allegheny Technologies
Anaergia
Anglo American Platinum
Arch Resources
Ardagh Group
Asahi
Axalta Coating Systems
Barrick Gold
BASF
BHP Group
Cabot
Carpenter Technology
Catcher Technology
Celanese
CF Industries
Commercial Metals
Compass Minerals International
CONSOL Energy
CRH
Danimer Scientific
Daqo New Energy
DSM
Ecolab
Eldorado Gold
Element Solutions
Elementis
FMC
Forterra
Fortescue Metals Group
Glencore
Gold Fields
Hexpol
Holcim
Huntsman
Ingevity
International Paper
Kaiser Aluminum
Kemira
Kinross Gold
LG Chem
Linde
Livent
Methanex
Nippon Carbon
Norsk Hydro
O-I Glass
Pactiv Evergreen
Perimeter Solutions
PPG Industries
PTT Global Chemical
RAO Norilsk Nickel
Reliance Steel & Aluminum
Rio Tinto
Sappi
Sasol
Sealed Air
Smurfit Kappa Group
Steel Dynamics
Suzano
Vale
W.R. Grace & Company
Wacker Chemie
WRKCo
Zurn Water Solutions

REAL ESTATE

Alexander & Baldwin
American Tower
Americold Realty
Annaly Capital Management
Assura
Boston Properties
British Land
Digital Realty Trust
Douglas Emmett
Empire State Realty Trust
Equinix
Equity LifeStyle Properties
Extra Space Storage
Goodman Group
JBG SMITH Properties
Life Storage
Mid-America Apartment Communities
Prologis
PS Business Parks
Public Storage
Segro
STORE Capital
Sun Communities
UNITE Group
VICI Properties
Welltower
Weyerhaeuser

UTILITIES

AES Corp
AB Ignitis Grupe
AES Andes
AES
Alliant Energy
American Electric Power
Array Technologies
Avangrid
Ballard Power Systems
Black Hills
CenterPoint Energy
CGN Power
Chubu Electric Power
Cia de Saneamento do Parana
Consolidated Edison
Dominion Energy
DTE Energy
Duke Energy
Dynegy
E.ON
Edison International
EDP - Energias de Portugal
Enel
Engie
ENN Energy
Entergy
Eversource Energy
Exelon
Iberdrola
Inter RAO
Israel Electric
Korea Electric Power
Maxeon Solar Technologies
National Grid
NextEra Energy
NRG Energy
OGE Energy
Oglethorpe Power
Oklahoma Gas and Electric
Orsted
Pampa Energia
PG&E
Pinnacle West Capital
Portland General Electric
RWE
Sempra Energy
Shoals Technologies
Southern
Suez
Sunnova Energy International
Terna - Rete Elettrica Nazionale
Tokyo Electric Power
UGI
Veolia Environnement
Verbund
WEC Energy

The companies shown comprise a complete list of all 2021 meetings in which Wellington Management participated, where climate change — related to physical risks, transition risks, or both — was among the discussion topics. These companies are not representative of all of the securities purchased, sold, or recommended for clients. It should not be assumed an investment in the companies listed has or will be profitable. Actual holdings will vary for each client and there is no guarantee a particular client's account will hold any or all of the companies shown. This material is not intended to constitute investment advice or an offer to sell, or the solicitation of an offer to purchase shares or other securities.

Important disclosures

This report is prepared in alignment with the recommendations of the Task Force on Climate-related Financial Disclosures ('Task Force'). The Task Force has established recommendations for disclosing comparable, reliable, and decision-useful information about the risks and opportunities presented by climate change. The Task Force structured its recommendations around four core elements:

Governance — The organization's governance around climate-related risks and opportunities

Strategy — The actual and potential impacts of climate- related risks and opportunities on the organization's businesses, strategy, and financial planning where such information is material

Risk Management — The processes used by the organization to identify, assess, and manage climate-related risks

Metrics and Targets — The metrics and targets used to assess and manage relevant climate-related risks and opportunities where such information is material

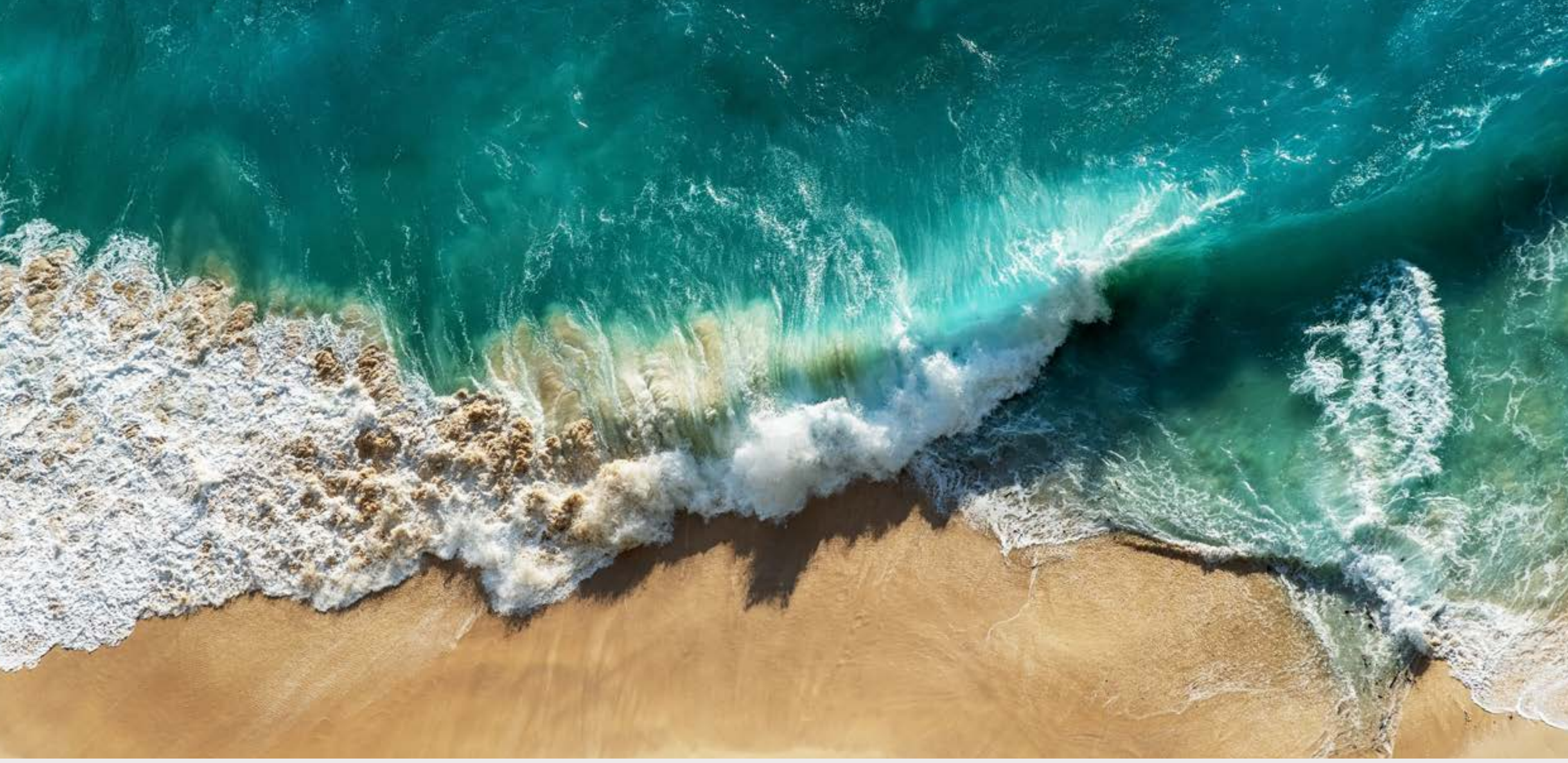
Certain data provided is that of a third party. While data is believed to be reliable, no assurance is being provided as to its accuracy or completeness. Company examples are for illustrative purposes only, are not representative of all investments made on behalf of our clients, and should not be interpreted as a recommendation or advice.

Individual portfolio management teams may hold different views and may make different investment decisions for different clients. Views may change over time. Each client account is individually managed; the extent to which climate-based research is considered in the investment process will vary depending on the investment policy established for each strategy or fund. Investors should always obtain and read an up-to-date investment services description or prospectus before deciding whether to appoint an investment manager or to invest in a fund.

Risks

All investments involve risks. Given the lengthy timeframes for most impact projects and many companies' reliance on disruptive technologies, investments may be subject to volatility and are therefore more suited to longer investment horizons. The following are some general risks associated with various approaches discussed in this report. This is not an all-inclusive list. Each specific investment approach and product will have its own specific risks and risks will vary.

Capital risk — The value of your investment may become worth more or less than at the time of the original investment. | **Concentration risk** — Concentration of investments in a relatively small number of securities, sectors or industries, or geographical regions may significantly affect performance. | **Equity and fixed income securities market risks** — Financial markets are subject to many factors, including economic conditions, government regulations, market sentiment, local and international political events, and environmental and technological issues. In addition, the market value of fixed income securities will fluctuate in response to changes in interest rates, currency values, and the creditworthiness of the issuer. | **Foreign and emerging markets risk** — Investments in foreign markets may present risks not typically associated with domestic markets. These risks may include changes in currency exchange rates; less-liquid markets and less available information; less government supervision of exchanges, brokers, and issuers; increased social, economic, and political uncertainty; and greater price volatility. These risks may be greater in emerging markets, which may also entail different risks from developed markets. | **Smaller-capitalization-stock risks** — The share prices of small- and mid-cap companies may exhibit greater volatility than the share prices of larger-capitalization companies. In addition, shares of small- and mid-cap companies are often less liquid than larger-cap companies. | **Manager risk** — Investment performance depends on the portfolio management team and the team's investment strategies. If the investment strategies do not perform as expected, if opportunities to implement those strategies do not arise, or if the team does not implement its investment strategies successfully, an investment portfolio may underperform or suffer significant losses. | **Sustainability risks** — ESG factors may be considered as part of broader analysis of individual issuers (including with regards to a sustainability risk assessment), using inputs from the investment manager's team of ESG analysts to help identify global best practices, prepare for engagement, and collaborate on new research inputs. The factors which will be considered will vary depending on the security in question, but typically include ownership structure, board structure and membership, capital allocation track record, management incentives, labor relations history, and climate risks.



IMPORTANT INFORMATION

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